

Siebert Relation Violation as a Probe of Spacetime Foam

A. G. Kim*

Physics Division, Lawrence Berkeley National Laboratory, Berkeley, California 94720, USA

P. E. Nugent

*Applied Mathematics and Computational Research Division,
Lawrence Berkeley National Laboratory, Berkeley, California 94720, USA*

L. Wang

Department of Physics and Astronomy, 4242 TAMU, College Station, Texas 77843-4242, USA

(Dated: May 21, 2026)

Quantum gravity predicts that spacetime fluctuates stochastically at the Planck scale — a “foam” that imprints a tiny achromatic arrival-time jitter on light from distant sources. We show that this jitter affects first- and second-order temporal coherence, $g^{(1)}$ and $g^{(2)}$, differently, producing a violation of the Siebert relation $g^{(2)}(\tau) - 1 = |g^{(1)}(\tau)|^2$ that otherwise holds for all Gaussian chaotic sources, when the foam fluctuates on timescales shorter than the optical coherence time. For an extended source the Siebert discrepancy is attenuated by an angular-size factor $\Phi_{\text{eff}}(r_{\perp}) \in [0, 1]$, so the point-source signal is the upper bound; the angular-size dependence itself provides a differential probe of the foam transverse coherence length $r_{\perp}$. Strongly lensed sources — whether persistent or transient — combine cosmological path length for maximum foam-variance accumulation with magnification-boosted photon rates, and represent the highest-priority targets when accessible. These observables provide constraints that are statistically competitive with and complementary to existing imaging tests, though exquisite control of systematic errors is required to access the full range of physically viable foam models.

I. INTRODUCTION

A long-standing expectation from the union of quantum mechanics and general relativity is that the smooth pseudo-Riemannian manifold of classical spacetime breaks down at the Planck length $\ell_P = \sqrt{\hbar G/c^3} \simeq 1.616 \times 10^{-35}$ m. In Wheeler’s original “spacetime foam” picture [1], quantum fluctuations of the metric on this scale render the geometry stochastic; similar conclusions are reached in the Euclidean path-integral treatment [2] and in modern holographic and entropic-gravity arguments from very different starting points. While a complete theory of quantum gravity remains elusive, the question of whether residual imprints of Planck-scale geometric fluctuations can accumulate over cosmological baselines and become observable is one of the few empirical handles on physics in the Planck regime, and it has motivated a substantial body of phenomenological work over the past two decades [3, 4].

A widely studied parametrization due to Ng and collaborators [4–6], the α -foam model, encodes the foam dynamics in a power-law path-length variance

$$\sigma_{\ell}^2(D) = A_{\alpha} D^{2(1-\alpha)} \ell_P^{2\alpha}, \quad (1)$$

where D is the proper distance to the source, α parametrizes the foam model, and A_{α} is a dimensionless coefficient of order unity. A photon traversing distance D passes through approximately $N \sim D/\ell_P$ independent Planck-scale spacetime cells, each imparting a tiny random displacement to the photon path; the exponent α controls how the per-cell displacements accumulate: for random-walk foam ($\alpha = 1/2$), contributions add in quadrature ($\sigma_{\ell} \propto \sqrt{D\ell_P}$); holographic foam ($\alpha = 2/3$) grows more slowly, constrained by an area-law bound on the number of independent degrees of freedom per Planck area; and conservative foam ($\alpha = 1$) yields a distance-independent $\sigma_{\ell} \propto \ell_P$, corresponding to perfectly correlated cell contributions.

The primary observational probe of stochastic foam to date has been the imaging channel. Spacetime foam accumulates a cumulative phase dispersion $\Delta\phi \simeq 2\pi\ell^{1-\alpha}\ell_P^{\alpha}/\lambda$ across the entrance aperture, degrading PSF sharpness and erasing interference fringes once $\Delta\phi \gtrsim 2\pi$ [7]. For PKS 1413+135 ($\ell = 1.216$ Gpc) observed with HST at $\lambda = 1.6 \mu\text{m}$, the random-walk model ($\alpha = 1/2$) predicts $\Delta\phi \sim 10 \times 2\pi$, so the detection of Airy rings in this image marginally rules out that model; the holographic model ($\alpha = 2/3$) predicts $\Delta\phi \sim 10^{-9} \times 2\pi$ and is obviously not ruled out [4, 8, 9]. Models with long-range transverse correlations evade even the random-walk bound [10].

* agkim@lbl.gov

We develop a framework based on the first- and second-order coherence functions $g^{(1)}(\tau)$ and $g^{(2)}(\tau)$ of light from astrophysical sources. The first-order coherence – the field–field autocorrelation, equivalently the fringe visibility at delay τ – inherits a foam suppression factor that decreases continuously from unity toward complete fringe erasure as the accumulated phase variance σ_ϕ^2 grows, extending the binary Lieu–Hillman erasure criterion [7] ($\sigma_\phi \gtrsim 1$) to a continuous curve and recovering the Strehl and fringe-visibility tests as limiting cases of the same statistic. The second-order coherence – the intensity–intensity autocorrelation measured in a Hanbury Brown–Twiss experiment [11] – is to leading order blind to the carrier-phase modulation that suppresses $g^{(1)}$, but acquires a distinct signature from the differential timing jitter between photon arrivals. The two together violate the Siegert relation $g^{(2)}(\tau) - 1 = |g^{(1)}(\tau)|^2$ [12] in a foam-specific way, providing a discriminant that does not require the intrinsic source visibility to be modeled. The finite-lag Siegert violation requires the foam to fluctuate on timescales shorter than the optical coherence time; in the quasi-static limit $\tau_{\text{foam}} \rightarrow \infty$ the signal vanishes. For an extended source the Siegert discrepancy is further attenuated by an angular-size factor $\Phi_{\text{eff}}(r_\perp) \in [0, 1]$, with the point-source limit ($\Phi_{\text{eff}} = 1$) saturating the upper bound; the source-size dependence is itself a discriminant for the foam transverse coherence length $r_\perp$.

Intuitively, $g^{(1)}$ measures the phase modulation of the electric field by the foam. The second-order coherence $g^{(2)}$ measures the differential arrival-time jitter between two photons. If both photons traverse identical foam fluctuations, their relative timing is unchanged and $g^{(2)}$ is unaffected. A change in $g^{(2)}$ requires the two photons to experience *different* foam: typically they follow the same path but are separated in time by more than the foam coherence time τ_{foam} , so the foam has changed between their passages. When the two photons arrive from sky directions separated enough to experience different foam fluctuations, $g^{(2)}$ is further suppressed.

The framework draws on a well-established phenomenology of dynamic phase screens acting on chaotic light. A phase screen that fluctuates within the detection timing window imprints intensity correlations on $g^{(2)}(\tau)$, the principle exploited in Martienssen & Spiller’s pseudo-thermal light source [13] and a recurring theme in HBT-through-turbulence studies [14, 15]. Planck-scale foam acts as a dynamic phase screen distributed along the entire line of sight: when $\tau_{\text{foam}} \lesssim \tau_L$ it produces a finite-lag Siegert discrepancy on point-like targets, attenuated by Φ_{eff} for sources whose angular extent resolves the foam transverse coherence scale $r_\perp$. The present work extends this laboratory phenomenon — dynamic-screen imprinting on $g^{(2)}$ — to cosmological propagation through spacetime foam.

Next-generation sensitivity is achievable via two complementary paths: large-aperture single telescopes measuring the HBT effect, and multi-telescope intensity interferometer arrays such as VERITAS [16, 17] — both increase collecting area and improve signal-to-noise for nearby sources, while the temporal $g^{(1)}$ channel is accessible to high-resolution spectroscopy. A detection campaign across targets of varying angular size would jointly probe $\{\alpha, r_\perp\}$ through the angular-size dependence of the Siegert discrepancy, providing a cross-check against common instrumental systematics.

The remainder of the paper is organized as follows. Section II develops the model: the spatiotemporal covariance of the foam-induced path-length fluctuation field, the differential-jitter distribution at a single ray, and its generalization to ray pairs. Section III develops the full coherence framework for point sources: first-order coherence and its closed-form results for Lorentzian and Voigt line profiles; second-order coherence and its narrow-band insensitivity to carrier phase; the resulting Siegert violation and its canonical detection observable; and the connection to the published Strehl and fringe-visibility tests. Section IV extends the analysis to spatially resolved sources, derives the angular-size suppression of the Siegert discrepancy by Φ_{eff} , shows that no foam-induced zero-lag bunching suppression arises in this model, and identifies distance scaling and achromaticity as discriminants against conventional propagation phase screens. Section V assesses detectability via single-aperture HBT measurements, derives aperture and signal-to-noise scaling, and surveys candidate astrophysical targets. Section VI summarizes the main results. The corrected $g_{\text{obs}}^{(2)}$ derivation underlying §IV is given in Appendix A.

II. SPACETIME FOAM MODEL

A. Arrival-Time Jitter as a Time Reparameterization

We model the detected field as a stochastically time-warped version of the intrinsic source field:

$$E_{\text{obs}}(t) = E_{\text{src}}(t - \delta t(t)), \quad (2)$$

where $E_{\text{src}}(t)$ is a single spatial mode of a stationary zero-mean complex Gaussian process (chaotic light, appropriate for virtually all known astrophysical sources) and $\delta t(t)$ is the foam-induced arrival-time jitter, a stationary stochastic process statistically independent of E_{src} . The jitter arises from cumulative path-length fluctuations along the line of sight of propagation distance D :

$$\delta t(t) = \frac{1}{c} \Delta \ell_{\text{tot}}(t). \quad (3)$$

In the geometric-optics limit the path-length shift $\Delta\ell_{\text{tot}}$ is a purely geometric quantity and hence frequency-independent (achromatic), in contrast to plasma dispersion which scales as λ^2 . This achromaticity is a key discriminant between foam and conventional propagation effects.

B. Spatial and Temporal Fluctuations of Spacetime Foam

The scalar variance $\sigma_\ell^2(D)$ (Eq. (1)) characterizes the amplitude of foam-induced path-length fluctuations along a single ray; here we describe how those fluctuations are correlated across ray pairs (spatial coherence) and across time (temporal decorrelation), both of which enter the HBT observables (see also Ref. [3] for a review of foam models and observational constraints).

Because foam is a property of spacetime rather than of any preferred direction, the same dimensional logic that governs longitudinal path-length fluctuations motivates the existence of a characteristic transverse coherence length $r_\perp$. However, the longitudinal and transverse cases are geometrically distinct: in the longitudinal case D counts the number of cells traversed along the ray, while in the transverse case b is a separation between rays that share cells along the line of sight — the number of shared cells depends on both b and D in a way that is not captured by a simple dimensional analogy. Accordingly, $r_\perp$ is treated as an independent free parameter in Section IV. We denote the normalized transverse coherence function by $\Phi(b/r_\perp)$, satisfying $\Phi(0) = 1$ and $\Phi(b/r_\perp) \rightarrow 0$ for $b \gg r_\perp$. The functional form of Φ depends on the specific foam model. For the pixellon model, Lee, Zurek & Chen [10] find that the transverse correlation structure involves long-range angular correlations (a Legendre polynomial series in aperture-plane angle) that do not reduce to a simple $\Phi(b/r_\perp)$ ansatz, but whose long-range character strongly suppresses image blurring relative to the uncorrelated case. In the present framework Φ is treated as a free function; a foam signal would manifest as an angular-size dependence of $\mathcal{S}(\tau)$ through the suppression factor Φ_{eff} (Section IV).

Turning to the temporal evolution of spacetime foam — an aspect not specified by Eq. (1) — quantum foam models posit that each Planck-scale cell along the line of sight fluctuates in time. Two photons traversing the same cell at coordinate times separated by τ experience path-length fluctuations that are correlated for $\tau \ll \tau_{\text{foam}}$ but decorrelated for $\tau \gg \tau_{\text{foam}}$, where τ_{foam} is the characteristic temporal fluctuation timescale at each cell. We introduce $C_\alpha(\tau; D)$ as the normalized *temporal* autocorrelation of the per-cell path-length fluctuation, integrated over the $\sim D/\ell_P$ cells along the line of sight. The timescale τ_{foam} is an additional free parameter of the foam model, not fixed by the α -model variance relation alone. If τ_{foam} is of order the Planck time, $\ell_P/c \approx 5 \times 10^{-44}$ s, then $C_\alpha(\tau) \approx 0$ for all accessible lags: the foam is fully decorrelated on any observable timescale, the most optimistic scenario for detection. In the opposite *quasi-static* limit $\tau_{\text{foam}} \gg D/c$ (foam coherent across the entire line of sight), $C_\alpha(\tau) \approx 1$ for all τ of interest and every C_α -dependent temporal observable vanishes: the finite-lag Siegert discrepancy $\mathcal{S}(\tau)$ is suppressed by $[1 - C_\alpha(\tau)] \rightarrow 0$ (Eq. (30)), and the single-ray $g_{\text{obs}}^{(1)}$ recovers its intrinsic value because the foam suppression factor $\mathcal{F}_{\text{foam}} = e^{-\sigma_\phi^2[1-C_\alpha]} \rightarrow 1$ (Eq. (23)). In this quasi-static limit the HBT framework loses sensitivity altogether and the foam manifests only through the imaging-channel $g^{(1)}$ suppression on time-averaged exposures. The full spatiotemporal covariance between two rays separated by transverse distance b takes the factorized form

$$\langle \Delta\ell_{\text{tot}}(t; \boldsymbol{\theta}) \Delta\ell_{\text{tot}}(t + \tau; \boldsymbol{\theta}') \rangle = \sigma_\ell^2(D) C_\alpha(\tau; D) \Phi\left(\frac{b(\boldsymbol{\theta}, \boldsymbol{\theta}')}{r_\perp}\right), \quad (4)$$

where $b(\boldsymbol{\theta}, \boldsymbol{\theta}') \simeq D|\boldsymbol{\theta} - \boldsymbol{\theta}'|$. The single-ray temporal covariance appropriate for a point source is the special case $\boldsymbol{\theta} = \boldsymbol{\theta}'$ (so $b = 0$):

$$\langle \Delta\ell_{\text{tot}}(t) \Delta\ell_{\text{tot}}(t + \tau) \rangle = \sigma_\ell^2(D) C_\alpha(\tau; D). \quad (5)$$

This product form — separating the total variance from the normalized spatiotemporal shape — is a structural assumption, not a consequence of Eq. (1), that defines a tractable class of foam models within which the analytic framework is exact. The parameters A_α , $C_\alpha(\tau; D)$, τ_{foam} , $r_\perp$, and Φ_{eff} parameterize distinct physical effects — single-ray variance, temporal foam dynamics, transverse coherence, and source angular structure — and each governs a distinct aspect of the signal's detectability. The factorization is consistent with phenomenologies in which the longitudinal path-length variance, the temporal decorrelation, and the transverse coherence arise from statistically independent mechanisms — e.g. the simple cellular pictures of Refs. [4–6] where each Planck-scale cell contributes an independent random kick.

The factorization is *not* generic: the pixellon model of Lee, Zurek & Chen [10] has long-range angular correlations whose spatiotemporal covariance does not reduce cleanly to $\sigma_\ell^2(D) C_\alpha(\tau) \Phi(b/r_\perp)$; the framework here applies to that model only as a heuristic approximation in the regime where transverse correlations have decoupled.

Observationally, factorization predicts that the extended-source Siegert discrepancy is suppressed relative to its point-source value by the angular-size factor $\Phi_{\text{eff}}(r_\perp)$: $\mathcal{S}_{\text{ext}}(\tau) \approx \Phi_{\text{eff}} \mathcal{S}_{\text{pt}}(\tau)$ in the weak-foam regime (Section IV),

independent of both τ and D . A dependence of this ratio on lag or source redshift would falsify the factorization ansatz and indicate that a non-factorizing model (e.g. pixellon) is required. The single observable developed below — the Siegert discrepancy $\mathcal{S}(\tau)$ — descends from Eq. (1), with Sections III–III F probing its longitudinal projection (single-ray temporal jitter) and Section IV its transverse projection (cross-ray Φ_{eff} suppression).

C. Differential Jitter and Its Distribution

With the spatiotemporal covariance model in place, we define the key derived quantity that enters the coherence integrals. Define the *cross-ray differential jitter* between a photon arriving along direction $\boldsymbol{\theta}$ at time $t + \tau$ and a photon arriving along $\boldsymbol{\theta}'$ at time t :

$$\Delta_{\boldsymbol{\theta}, \boldsymbol{\theta}'}(\tau) \equiv \delta t(t + \tau; \boldsymbol{\theta}) - \delta t(t; \boldsymbol{\theta}'). \quad (6)$$

For fixed D , stationarity implies the statistics depend only on τ and the transverse separation $b(\boldsymbol{\theta}, \boldsymbol{\theta}') = D|\boldsymbol{\theta} - \boldsymbol{\theta}'|$. Expanding the square and applying Eq. (4):

$$\begin{aligned} \sigma_{\Delta}^2(\tau; \boldsymbol{\theta}, \boldsymbol{\theta}') &= \frac{1}{c^2} \langle [\Delta \ell_{\text{tot}}(t + \tau; \boldsymbol{\theta}) - \Delta \ell_{\text{tot}}(t; \boldsymbol{\theta}')]^2 \rangle \\ &= \frac{1}{c^2} \left[\sigma_{\ell}^2 + \sigma_{\ell}^2 - 2\sigma_{\ell}^2 C_{\alpha}(\tau) \Phi\left(\frac{b}{r_{\perp}}\right) \right] \\ &= \frac{2\sigma_{\ell}^2(D)}{c^2} \left[1 - C_{\alpha}(\tau; D) \Phi\left(\frac{b(\boldsymbol{\theta}, \boldsymbol{\theta}')}{r_{\perp}}\right) \right]. \end{aligned} \quad (7)$$

For $b \gg r_{\perp}$, $\Phi \rightarrow 0$ and the variance becomes $2\sigma_{\ell}^2/c^2$ at all lags, including $\tau = 0$ — a new effect absent in the point-source case, and one that is independent of the temporal fluctuation structure of the foam (i.e. independent of C_{α}). In the dynamical limit $C_{\alpha} \rightarrow 0$ this result holds for all separations b , independently of Φ .

By the central limit theorem (CLT), $\Delta_{\boldsymbol{\theta}, \boldsymbol{\theta}'}(\tau)$ is a sum of contributions from many foam cells and is therefore Gaussian with zero mean:

$$\Delta_{\boldsymbol{\theta}, \boldsymbol{\theta}'}(\tau) \sim \mathcal{N}(0, \sigma_{\Delta}^2(\tau; \boldsymbol{\theta}, \boldsymbol{\theta}')), \quad (8)$$

where σ_{Δ}^2 is taken from Eq. (7).

The Gaussian approximation follows from the CLT: for any source at $D \gtrsim 1$ kpc, the cell count $N_{\text{cells}} \sim D/\ell_{\text{P}} \gtrsim 10^{46}$ amply satisfies the large- N requirement and higher-cumulant corrections are negligible. For the cross-ray case the additional condition $b \gg r_{\perp}$ is assumed, and is satisfied in the fully-resolved limit $\Phi_{\text{eff}} \rightarrow 0$ adopted in Section V.

III. COHERENCE FUNCTIONS AND SIEGERT VIOLATION

A. Point-Source Differential Jitter

In this section we restrict to a point source ($b = 0$), for which the cross-ray differential jitter of Eq. (6) reduces to the single-ray differential jitter

$$\Delta(\tau) \equiv \delta t(t + \tau) - \delta t(t), \quad (9)$$

with variance obtained from Eq. (7) at $\Phi(0) = 1$:

$$\sigma_{\Delta}^2(\tau; D) = \frac{2\sigma_{\ell}^2(D)}{c^2} [1 - C_{\alpha}(\tau; D)]. \quad (10)$$

Although this section restricts to a point source for notational compactness, this single-ray variance applies equally to extended sources at finite lag $\tau \neq 0$: under the factorized covariance Eq. (4), the temporal factor $C_{\alpha}(\tau; D)$ is identical for all ray pairs, so cross-ray transverse structure enters only at $\tau = 0$ through Φ_{eff} (Section IV).

B. Jitter Convolution Formula for $g^{(1)}$

The observed first-order coherence is

$$g_{\text{obs}}^{(1)}(\tau) = \frac{\langle E_{\text{obs}}^*(t) E_{\text{obs}}(t + \tau) \rangle}{\langle |E_{\text{obs}}(t)|^2 \rangle}. \quad (11)$$

Substituting Eq. (2) and letting $u = t - \delta t(t)$, $v = t + \tau - \delta t(t + \tau)$, so that $v - u = \tau - \Delta(\tau)$, stationarity of E_{src} gives

$$\langle E_{\text{src}}^*(u) E_{\text{src}}(v) \rangle = \langle |E_{\text{src}}|^2 \rangle g_{\text{src}}^{(1)}(\tau - \Delta(\tau)). \quad (12)$$

Since δt is independent of E_{src} , averaging over the jitter yields

$$g_{\text{obs}}^{(1)}(\tau) = \langle g_{\text{src}}^{(1)}(\tau - \Delta(\tau)) \rangle_{\delta t} = \int_{-\infty}^{\infty} \mathcal{N}(\Delta; 0, \sigma_{\Delta}^2(\tau)) g_{\text{src}}^{(1)}(\tau - \Delta) d\Delta. \quad (13)$$

The observed first-order coherence is the intrinsic coherence function convolved with the Gaussian differential-jitter kernel.

C. Closed Forms for Lorentzian and Voigt Sources

As a representative case admitting a closed-form result, we take a Lorentzian line profile, which corresponds to an exponential coherence function via the Wiener–Khinchin theorem:

$$g_{\text{src}}^{(1)}(\tau) = e^{-|\tau|/\tau_L}, \quad (14)$$

where τ_L is the coherence time. Evaluating Eq. (13) by completing the square and splitting on the sign of $\tau - \Delta$ gives, for $\tau > 0$:

$$g_{\text{obs}}^{(1)}(\tau) = e^{\sigma_{\Delta}^2/2\tau_L^2} \cdot \frac{1}{2} \left[e^{-\tau/\tau_L} \operatorname{erfc}\left(\frac{\sigma_{\Delta}/\tau_L - \tau/\sigma_{\Delta}}{\sqrt{2}}\right) + e^{\tau/\tau_L} \operatorname{erfc}\left(\frac{\sigma_{\Delta}/\tau_L + \tau/\sigma_{\Delta}}{\sqrt{2}}\right) \right]. \quad (15)$$

In the weak-jitter limit $\sigma_{\Delta} \ll \tau_L$, the Gaussian average of $e^{-|\tau - \Delta|/\tau_L}$ over $\Delta \sim \mathcal{N}(0, \sigma_{\Delta}^2)$ is dominated by $|\Delta| \ll |\tau|$, so one expands around $\Delta = 0$:

$$g_{\text{obs}}^{(1)}(\tau) \approx e^{-|\tau|/\tau_L} \left[1 + \frac{\sigma_{\Delta}^2(\tau)}{2\tau_L^2} + \mathcal{O}(\sigma_{\Delta}^4) \right]. \quad (16)$$

Here $\sigma_{\Delta}(\tau)$ is τ -dependent through $C_{\alpha}(\tau)$, and the expansion is in $\sigma_{\Delta}(\tau)/\tau_L$ at fixed τ . The correction is positive because $e^{-|\tau|/\tau_L}$ is convex and Jensen's inequality applies. The normalization $g_{\text{obs}}^{(1)}(0) = 1$ is preserved exactly.

For a Voigt profile $g_{\text{src}}^{(1)}(\tau) = e^{-|\tau|/\tau_L} e^{-\tau^2/2\tau_G^2}$ — a Lorentzian of width τ_L (from optically-thick electron scattering or BLR turbulent broadening) convolved with a Gaussian of width τ_G (from bulk Doppler motions) — the jitter kernel combines with the source Gaussian into an effective width and mean,

$$\tilde{\sigma}_1^2 = \frac{\sigma_{\Delta}^2 \tau_G^2}{\sigma_{\Delta}^2 + \tau_G^2}, \quad \mu_1(\tau) = \frac{\tau_G^2 \tau}{\sigma_{\Delta}^2 + \tau_G^2}, \quad (17)$$

giving, for $\tau > 0$:

$$g_{\text{obs}}^{(1)}(\tau) = \frac{\tilde{\sigma}_1}{\sigma_{\Delta}} e^{-\tau^2/2(\sigma_{\Delta}^2 + \tau_G^2)} e^{\tilde{\sigma}_1^2/2\tau_L^2} \frac{1}{2} \left[e^{-\mu_1/\tau_L} \operatorname{erfc}\left(\frac{\tilde{\sigma}_1/\tau_L - \mu_1/\tilde{\sigma}_1}{\sqrt{2}}\right) + e^{\mu_1/\tau_L} \operatorname{erfc}\left(\frac{\tilde{\sigma}_1/\tau_L + \mu_1/\tilde{\sigma}_1}{\sqrt{2}}\right) \right]. \quad (18)$$

The normalization $g_{\text{obs}}^{(1)}(0) = 1$ is preserved: recall $\sigma_{\Delta} = \sigma_{\Delta}(\tau)$ with $\sigma_{\Delta}(0) = 0$ (Eq. (10)), so $\tilde{\sigma}_1(0) = 0$, the exponential prefactors approach unity, and the erfc terms reduce to $\operatorname{erfc}(0) + \operatorname{erfc}(0) = 2$. In the limit $\tau_G \rightarrow \infty$ (pure Lorentzian), $\tilde{\sigma}_1 \rightarrow \sigma_{\Delta}$ and $\mu_1 \rightarrow \tau$, so Eq. (18) reduces to Eq. (15). In the limit $\tau_L \rightarrow \infty$ (pure Gaussian), the erfc terms combine to unity and $g_{\text{obs}}^{(1)}(\tau) = (\tilde{\sigma}_1/\sigma_{\Delta}) e^{-\tau^2/2(\sigma_{\Delta}^2 + \tau_G^2)}$: the observed coherence is a Gaussian whose width is broadened in quadrature by the jitter.

D. Narrow-Band Limit

For a quasi-monochromatic source with center frequency ω_0 and angular-frequency bandwidth $\Delta\omega \ll \omega_0$, $E_{\text{src}}(t) = A(t) e^{-i\omega_0 t}$ where $A(t)$ varies on the coherence timescale $\tau_c = 1/\Delta\omega = \lambda^2/(2\pi c \Delta\lambda)$. When $\sigma_\ell/(c\tau_c) \ll 1$, typical realizations of δt leave the envelope unaffected, $A(t - \delta t) \approx A(t)$, and the jitter acts purely as a carrier phase modulation; the observed field takes the form

$$E_{\text{obs}}(t) \approx A(t) e^{-i\omega_0 t} e^{i\omega_0 \delta t(t)}. \quad (19)$$

(Corrections to the envelope approximation are of order $(\sigma_\ell/c\tau_c)^2$; for $\alpha = 1/2$ at $D = 1$ Gpc, $\lambda = 500$ nm, $\Delta\lambda/\lambda = 10^{-3}$ this parameter is ≈ 0.28 , so the exact convolution Eq. (15) should be used instead. For $\alpha = 2/3$ the analogous parameter is $\sim 10^{-11}$ (Section III F 2), so Eq. (21) is exact to any practical precision in the holographic case.) Since $A(t)$ and $\delta t(t)$ are independent, the first-order coherence factorizes. The jitter contribution is the characteristic function of $\Delta(\tau)$ evaluated at ω_0 ; since $\Delta(\tau)$ is Gaussian (Section II C),

$$\langle e^{i\omega_0 \Delta(\tau)} \rangle_{\delta t} = e^{-\omega_0^2 \sigma_\Delta^2(\tau)/2}. \quad (20)$$

This holds when $\tau \gg \tau_{\text{foam}}$ so that foam-cell contributions are independent; in the dynamical limit $\tau_{\text{foam}} \ll \tau_c$ the condition is satisfied at all observationally relevant lags. Substituting $\omega_0 = 2\pi c/\lambda$ and Eq. (10):

$$g_{\text{obs}}^{(1)}(\tau) = g_{\text{src}}^{(1)}(\tau) \cdot \exp(-\sigma_\phi^2 [1 - C_\alpha(\tau; D)]), \quad (21)$$

where we define the rms accumulated phase

$$\sigma_\phi^2 \equiv \left(\frac{2\pi}{\lambda_{\text{obs}}} \right)^2 \int_0^L [1 + z(s)]^2 \frac{d\sigma_\ell^2}{ds} ds, \quad (22)$$

integrating the foam differential variance $d\sigma_\ell^2/ds = A_\alpha 2(1 - \alpha) s^{1-2\alpha} \ell_P^{2\alpha}$ along the photon's local proper path of total length L . The $(1 + z)^2$ weighting reflects that a foam cell at redshift z imparts an observer-frame time delay $\delta t_{\text{obs}} = (1 + z) \delta \ell_{\text{local}}/c$, and hence a phase $\delta\phi = (2\pi/\lambda_{\text{obs}})(1 + z) \delta \ell_{\text{local}}$, because the photon's wavelength at z is $\lambda_{\text{local}}(z) = \lambda_{\text{obs}}/(1 + z)$. Switching integration variable to comoving distance via $ds = dr_C/(1 + z)$ yields the equivalent form $\sigma_\phi^2 = (2\pi/\lambda_{\text{obs}})^2 \int_0^{D_C} (1 + z) (d\sigma_\ell^2/ds) dr_C$. For $z \ll 1$ both forms reduce to the flat-space limit $\sigma_\phi = 2\pi \sqrt{A_\alpha} D^{1-\alpha} \ell_P^\alpha/\lambda$. We write the foam suppression factor as

$$\mathcal{F}_{\text{foam}}(\tau) \equiv e^{-\sigma_\phi^2 [1 - C_\alpha(\tau; D)]}, \quad (23)$$

so $g_{\text{obs}}^{(1)}(\tau) = g_{\text{src}}^{(1)}(\tau) \cdot \mathcal{F}_{\text{foam}}(\tau)$. At $\tau = 0$: $\mathcal{F}_{\text{foam}}(0) = 1$. At $\tau \rightarrow \infty$: $\mathcal{F}_{\text{foam}}(\infty) = e^{-\sigma_\phi^2 [1 - C_\alpha(\infty; D)]}$, which equals $e^{-\sigma_\phi^2}$ in the dynamical limit ($C_\alpha \rightarrow 0$) and approaches unity for quasi-static foam ($C_\alpha \approx 1$). The narrow-band condition $\sigma_\ell/(c\tau_c) \ll 1$ is compatible with the weak-jitter expansion of Eq. (16) — the former controls envelope displacement, the latter the carrier-phase suppression — and when neither applies the full convolution Eq. (13) must be evaluated.

E. Second-Order Coherence

We derive the second-order coherence $g_{\text{obs}}^{(2)}(\tau)$ for a point source, continuing the framework of Section III. The result has the same structure as Eq. (13): $g_{\text{obs}}^{(2)} - 1$ is a convolution of $g_{\text{src}}^{(2)} - 1$ with the identical Gaussian jitter kernel. Because the intensity $|E_{\text{obs}}|^2$ is blind to the carrier phase $e^{i\omega_0 \delta t}$, foam leaves $g^{(2)}$ nearly unchanged in the narrow-band limit while strongly suppressing $g^{(1)}$ — the asymmetry that produces the Siegert violation developed in Section III F.

The same argument makes $g^{(2)}$ practically immune to atmospheric seeing under current observing conditions, in contrast to $g^{(1)}$. Atmospheric turbulence introduces path-length changes of order $\Delta \ell_{\text{atm}} \sim 0.1\text{--}1$ mm, far exceeding the optical wavelength; this randomizes the carrier phase and destroys fringe visibility without adaptive optics. However, $g^{(2)}$ is blind to this carrier-phase modulation, and the atmospheric coherence time $\tau_0 \sim 1$ ms far exceeds the HBT correlation window $\tau_L \sim$ ns, so both photons in a coincidence traverse a frozen atmosphere and the differential atmospheric jitter is negligible. This advantage narrows as detector timing resolution approaches the atmospheric timing jitter $\Delta \ell_{\text{atm}}/c \sim 1\text{--}3$ ps; sub-ps detectors will require atmospheric monitoring or site-selection to control this systematic.

The Wick-contraction derivation of Appendix A, specialized to a point source ($\Phi = 1$, $X = Y \equiv \Delta(\tau)$), yields

$$g_{\text{obs}}^{(2)}(\tau) - 1 = \int_{-\infty}^{\infty} \mathcal{N}(\Delta; 0, \sigma_\Delta^2(\tau)) [g_{\text{src}}^{(2)}(\tau - \Delta) - 1] d\Delta, \quad (24)$$

with $\sigma_\Delta^2(\tau) = (2\sigma_\ell^2(D)/c^2)[1 - C_\alpha(\tau; D)]$ the same single-ray differential variance that enters Eq. (13). The structure is the same as the $g^{(1)}$ result: a Gaussian jitter kernel convolves the intrinsic source coherence. The Gaussian kernel requires $\tau \gg \tau_{\text{foam}}$; for $\tau \lesssim \tau_{\text{foam}}$ Eq. (24) should be regarded as approximate (Section II B).

The intensity $I_{\text{obs}}(t) = |E_{\text{obs}}(t)|^2$ is blind to the carrier phase $e^{i\omega_0\delta t(t)}$: only the envelope matters. In the narrow-band limit $\delta t \ll \tau_c$, the jitter does not appreciably shift the envelope, $A(t - \delta t) \approx A(t)$, so $I_{\text{obs}}(t) \approx |A(t)|^2 = I_{\text{src}}(t)$ to leading order in $\delta t/\tau_c$. Consequently:

$$g_{\text{obs}}^{(2)}(\tau) = g_{\text{src}}^{(2)}(\tau) \quad (\text{leading order in } \delta t/\tau_c), \quad (25)$$

with corrections of order $(\sigma_\ell/c\tau_c)^2$ from both envelope-timing and carrier-phase contributions to the intensity. For $\alpha = 1/2$ at $D = 1$ Gpc, $\lambda = 500$ nm, $\Delta\lambda/\lambda = 10^{-3}$: $\delta t_{\text{rms}}/\tau_c \approx 0.28$, so corrections are at the $\sim 8\%$ level and the exact convolution Eq. (24) should be used for precision work. For $\alpha = 2/3$ the expansion parameter $\sigma_\ell/(c\tau_c)$ is $\sim 10^{-11}$ and Eq. (25) is exact to any practical precision.

By contrast, $g^{(1)}$ is sensitive to the carrier phase because $\sigma_\phi = \omega_0\sigma_\ell/c$ can be large even when $\delta t \ll \tau_c$: for the same parameters, $\sigma_\phi \approx 280 \gg 1$.

F. Siegert Relation Violation

The Siegert relation $g^{(2)}(\tau) - 1 = |g^{(1)}(\tau)|^2$ holds for any stationary complex Gaussian field and provides a powerful internal consistency check: once $g^{(1)}$ is known, the expected $g^{(2)}$ is fixed. As foreshadowed in Section III E, foam violates this relation because $g_{\text{obs}}^{(1)}$ is suppressed by the carrier-phase factor $\mathcal{F}_{\text{foam}}$ while $g_{\text{obs}}^{(2)}$ is not. This subsection derives and characterizes the Siegert discrepancy $\mathcal{S}(\tau) \equiv [g_{\text{obs}}^{(2)} - 1] - |g_{\text{obs}}^{(1)}|^2$.

1. General Bound

Define the Siegert discrepancy:

$$\mathcal{S}(\tau) \equiv [g_{\text{obs}}^{(2)}(\tau) - 1] - |g_{\text{obs}}^{(1)}(\tau)|^2. \quad (26)$$

Let $X \equiv g_{\text{src}}^{(1)}(\tau - \Delta(\tau))$, a complex random variable. From Eqs. (13) and (24):

$$g_{\text{obs}}^{(1)}(\tau) = \langle X \rangle_{\delta t}, \quad g_{\text{obs}}^{(2)}(\tau) - 1 = \langle |X|^2 \rangle_{\delta t}. \quad (27)$$

Therefore

$$\mathcal{S}(\tau) = \langle |X|^2 \rangle - |\langle X \rangle|^2 \geq 0, \quad (28)$$

by the Cauchy–Schwarz inequality, with equality only when $\Delta(\tau)$ has zero variance (i.e., at $\tau = 0$ or in the absence of foam). The Siegert relation is violated for all $\tau \neq 0$ whenever foam is present. The Siegert violation is not a non-classical signature: the foam-modified field remains a classical stochastic process, and the classical inequalities $g^{(2)}(0) \geq g^{(2)}(\tau)$ and $g^{(2)}(0) > 1$ are preserved; it is only the Gaussian-statistics identity relating $g^{(2)}$ to $g^{(1)}$ that is broken, due to the stochastic phase accumulated along the propagation path.

2. Sensitivity Scale Decoupling

The violation arises because $g^{(1)}$ and $g^{(2)}$ respond to foam on different scales. The first-order coherence is controlled by $\sigma_\phi = 2\pi\sigma_\ell(D)/\lambda$, the rms accumulated phase along a single ray; $g^{(1)}$ is sensitive to it because the carrier phase $e^{i\omega_0\delta t}$ appears directly in the field correlation. The second-order coherence is controlled by $\sigma_\Delta(\tau)/\tau_c$, the rms differential jitter scaled by the coherence time; $g^{(2)}$ is sensitive to it because intensity correlations depend on how much the jitter *changes* between two detection events, not its absolute value. These two scales are related by

$$\frac{\sigma_\Delta(\tau)}{\tau_c} = \frac{\sqrt{2[1 - C_\alpha(\tau)]} \sigma_\ell(D)}{c\tau_c} = \sigma_\phi \cdot \frac{\Delta\lambda}{\lambda} \cdot \sqrt{2[1 - C_\alpha(\tau)]}, \quad (29)$$

using $\omega_0\tau_c = \lambda/\Delta\lambda$. The factor $\Delta\lambda/\lambda \ll 1$ means that even when $\sigma_\phi \gg 1$ (fringes erased), $\sigma_\Delta(\tau)/\tau_c$ can be small — so for a *point source* $g^{(2)}$ remains near its intrinsic value while $g^{(1)}$ is suppressed. This decoupling is the physical

root of the Siegert violation. For an extended source the decoupling is incomplete: at finite lag $g_{\text{obs}}^{(2)}$ picks up the carrier-phase scale σ_ϕ through the cross-direction factor $\mathcal{F}_{\text{cross}}$ of Eq. (44), attenuating the Siegert discrepancy by the angular-size factor Φ_{eff} (Section IV). At the fiducial parameters $D = 1$ Gpc, $\lambda = 500$ nm, $\Delta\lambda/\lambda = 10^{-3}$, $A_\alpha = 1$, and the dynamical limit $C_\alpha \rightarrow 0$, the random-walk model $\alpha = 1/2$ gives $\sigma_\phi = 280$ and $\sigma_\Delta(\tau)/\tau_c = 0.40$, while the holographic model $\alpha = 2/3$ gives $\sigma_\phi = 2.5 \times 10^{-8}$ and $\sigma_\Delta(\tau)/\tau_c = 3.6 \times 10^{-11}$. For quasi-static foam ($C_\alpha \approx 1$) the ratio $\sigma_\Delta(\tau)/\tau_c$ vanishes at all lags. The $\alpha = 1/2$ case is excluded by imaging under the standard assumption of short-range transverse foam correlations [9]; models with long-range correlations evade this bound [10], and $\alpha = 2/3$ (holographic) is the most sensitive surviving model.

3. Explicit Form and Properties in the Narrow-Band Limit

In the narrow-band limit, $g_{\text{obs}}^{(2)} \approx g_{\text{src}}^{(2)}$ (Eq. (25)), while $g_{\text{obs}}^{(1)}$ is suppressed by $\mathcal{F}_{\text{foam}}(\tau)$ (Eq. (21)). Using the intrinsic Siegert relation $g_{\text{src}}^{(2)}(\tau) - 1 = |g_{\text{src}}^{(1)}(\tau)|^2$, to leading order:

$$\mathcal{S}(\tau) \approx |g_{\text{src}}^{(1)}(\tau)|^2 \left(1 - e^{-2\sigma_\phi^2[1-C_\alpha(\tau;D)]}\right). \quad (30)$$

In the weak-jitter limit $\sigma_\phi \ll 1$, $\mathcal{S} \approx 2\sigma_\phi^2[1-C_\alpha]|g_{\text{src}}^{(1)}|^2$ — linear in σ_ϕ^2 and small. In the strong-jitter limit $\sigma_\phi \gg 1$, the exponential vanishes and $\mathcal{S} \approx |g_{\text{src}}^{(1)}(\tau)|^2$: $g_{\text{obs}}^{(2)} - 1$ retains its intrinsic value while $|g_{\text{obs}}^{(1)}|^2 \rightarrow 0$, producing a full violation. Quasi-static foam ($C_\alpha \approx 1$) gives $\mathcal{S} \approx 0$ at all lags — foam is invisible to all coherence observables in this limit. The discrepancy $\mathcal{S}(\tau)$ is constructed directly from the two measurable quantities $|g_{\text{obs}}^{(1)}(\tau)|$ and $g_{\text{obs}}^{(2)}(\tau) - 1$ (or equivalently $\mathcal{S}(\nu)$ in the spectral domain), and the modulation beyond the prefactor $|g_{\text{src}}^{(1)}(\tau)|^2$ is controlled solely by α , D , and $C_\alpha(\tau;D)$, independently of the unknown source spectrum.

For a Lorentzian source, $\mathcal{S}(\tau)$ peaks at a lag set by the competition between the source-coherence falloff $e^{-2\tau/\tau_L}$ and the foam-induced rise $1 - e^{-2\sigma_\phi^2[1-C_\alpha(\tau)]}$. Differentiating Eq. (30) yields the implicit condition

$$-\sigma_\phi^2 C'_\alpha(\tau_{\text{max}}) e^{-2\sigma_\phi^2[1-C_\alpha(\tau_{\text{max}})]} = \frac{1}{\tau_L} \left(1 - e^{-2\sigma_\phi^2[1-C_\alpha(\tau_{\text{max}})]}\right). \quad (31)$$

In the weak-jitter limit ($\sigma_\phi^2 \ll 1$) this reduces to the σ_ϕ -independent statement $-C'_\alpha(\tau_{\text{max}})/[1-C_\alpha(\tau_{\text{max}})] = 2/\tau_L$. For the canonical exponential decorrelation $C_\alpha(\tau) = e^{-\tau/\tau_{\text{foam}}}$,

$$\tau_{\text{max}} = \tau_{\text{foam}} \ln\left(1 + \frac{\tau_L}{2\tau_{\text{foam}}}\right), \quad (32)$$

which interpolates between $\tau_{\text{max}} \approx \tau_{\text{foam}} \ln[\tau_L/(2\tau_{\text{foam}})]$ for fast foam ($\tau_{\text{foam}} \ll \tau_L$) and $\tau_{\text{max}} \rightarrow \tau_L/2$ for slow foam ($\tau_{\text{foam}} \gg \tau_L$).

In the fast-foam / dynamical limit ($\tau_{\text{foam}} \ll \tau_L$, $C_\alpha \rightarrow 0$ above τ_{foam}), $|g_{\text{src}}^{(1)}(\tau_{\text{max}})|^2 \rightarrow 1$ and $1 - C_\alpha(\tau_{\text{max}}) \rightarrow 1$, so Eq. (30) reduces to

$$\mathcal{S}(\tau_{\text{max}}) \approx 1 - e^{-2\sigma_\phi^2}, \quad (33)$$

the canonical detection observable: quadratic $\mathcal{S}(\tau_{\text{max}}) \approx 2\sigma_\phi^2$ in the weak-foam limit, saturating at 1 for $\sigma_\phi \gg 1$.

G. Connection to Fringe Visibility and Strehl Ratio

The fringe visibility and Strehl ratio are spatial-coherence observables — they probe phase variations across an aperture or between two separated telescopes, described by $g_{\text{obs}}^{(1)}(B)$ — while $g_{\text{obs}}^{(1)}(\tau)$ derived above describes temporal coherence along a single line of sight. The two cases are directly analogous: the temporal decorrelation $C_\alpha(\tau;D)$, measuring how the foam changes between two observations separated by time τ , maps to the spatial decorrelation $\Phi(B/r_\perp)$ (Section II), measuring how correlated the foam fluctuations are at two aperture points separated by baseline B . In both cases the foam suppression has the same form, with C_α replaced by Φ :

$$\mathcal{F}_{\text{foam}}^{\text{spat}}(B) \equiv e^{-\sigma_\phi^2[1-\Phi(B/r_\perp)]}. \quad (34)$$

This mirrors $\mathcal{F}_{\text{foam}}(\tau)$ exactly: $\Phi(0) = 1$ gives no suppression at zero baseline, while $\Phi(B/r_\perp) \rightarrow 0$ for $B \gg r_\perp$ gives maximum suppression $e^{-\sigma_\phi^2}$ — just as $\mathcal{F}_{\text{foam}}(0) = 1$ and $\mathcal{F}_{\text{foam}}(\infty) = e^{-\sigma_\phi^2}$.

The fringe visibility at baseline B is then

$$V(B) = V_{\text{src}}(B) \cdot \mathcal{F}_{\text{foam}}^{\text{spat}}(B) = V_{\text{src}}(B) \cdot e^{-\sigma_\phi^2[1-\Phi(B/r_\perp)]}, \quad (35)$$

where $V_{\text{src}}(B) = |g_{\text{src}}^{(1)}|$ evaluated at angular scale λ/B . In the limit $B \gg r_\perp$ the two aperture points sample independent foam realizations and the suppression reaches its maximum $e^{-\sigma_\phi^2}$. Lieu & Hillman [7] implicitly assumed this limit: their erasure condition $\Delta\phi \geq 2\pi$ with $\Delta\phi \simeq 2\pi\sigma_\ell(D)/\lambda$ [4] is precisely $\sigma_\phi \geq 1$, the limit $\mathcal{F}_{\text{foam}}^{\text{spat}} \rightarrow 0$. Equation (35) extends their binary criterion to a continuous suppression curve whose shape encodes $\Phi(B/r_\perp)$ — in direct analogy to how $g_{\text{obs}}^{(1)}(\tau)$ encodes the temporal decorrelation $C_\alpha(\tau; D)$.

The Strehl ratio for a single aperture of diameter D_{tel} follows the same logic with $B \sim D_{\text{tel}}$: when $r_\perp \ll D_{\text{tel}}$, aperture-plane phases are independent draws from the single-ray distribution and $S = e^{-\sigma_\phi^2}$. [18] Both Strehl and fringe visibility therefore measure the same underlying $\sigma_\phi^2 = (2\pi\sigma_\ell/\lambda)^2$ in the large-baseline limit $B \gg r_\perp$, and the published constraints [4, 7–9, 19] are unified within this framework.

IV. EXTENDED SOURCES: FINITE-LAG SUPPRESSION OF THE SIEGERT DISCREPANCY

We now generalize from a point source to an extended incoherent source. Three results emerge: (i) the first-order coherence $g_{\text{obs}}^{(1)}$ is unchanged from the point-source case; (ii) the second-order coherence at zero lag is exactly $g_{\text{obs}}^{(2)}(0) = 2$ — a single-mode detector observing chaotic light from an extended source develops no foam-induced suppression of the zero-lag bunching peak; and (iii) at finite lag the extended-source Siegert discrepancy is *suppressed* relative to the point-source value by an angular-size factor $\Phi_{\text{eff}} \in [0, 1]$, so the point-source Siegert discrepancy is the maximum-signal upper bound for the foam signature in this framework. The full pedestrian derivation is given in Appendix A; this section presents the key results.

A. Field Model and Spatiotemporal Covariance

We generalize to an extended incoherent source whose angular brightness distribution is $I(\boldsymbol{\theta})$ (normalized to unity). The analysis assumes a point (single spatial mode) detector; a finite aperture would introduce Van Cittert–Zernike geometric phases that modify the angular coherence. The detected field is a weighted superposition over rays labelled by direction offset $\boldsymbol{\theta}$:

$$E_{\text{obs}}(t) = \int d\boldsymbol{\theta} I^{1/2}(\boldsymbol{\theta}) E_{\boldsymbol{\theta}}(t - \delta t(t; \boldsymbol{\theta})), \quad (36)$$

where the $E_{\boldsymbol{\theta}}$ are independent, identically distributed complex Gaussian processes (one per sky direction), and each ray carries its own foam jitter $\delta t(t; \boldsymbol{\theta})$. The spatiotemporal covariance of the path-length aggregate is given by Eq. (4) of Section II, with $b(\boldsymbol{\theta}, \boldsymbol{\theta}') \simeq D|\boldsymbol{\theta} - \boldsymbol{\theta}'|$ the transverse separation between rays evaluated at the source end. For foam distributed uniformly along the line of sight, the ray separation varies as $z|\boldsymbol{\theta} - \boldsymbol{\theta}'|$; evaluating at the source plane $b = D|\boldsymbol{\theta} - \boldsymbol{\theta}'|$ introduces an $O(1)$ geometric factor that is absorbed into the free parameter $r_\perp$.

The first-order coherence is unchanged from the point-source case. Spatial incoherence ($\langle E_{\boldsymbol{\theta}}^* E_{\boldsymbol{\theta}'} \rangle \propto \delta(\boldsymbol{\theta} - \boldsymbol{\theta}')$) forces only the diagonal to contribute; the angular integral over I gives unity, so Eq. (21) holds without modification:

$$g_{\text{obs}}^{(1)}(\tau) = g_{\text{src}}^{(1)}(\tau) \cdot e^{-\sigma_\phi^2[1-C_\alpha(\tau; D)]}. \quad (37)$$

It is convenient to define the effective transverse coherence factor weighting $\Phi(b/r_\perp)$ by the source brightness distribution:

$$\Phi_{\text{eff}} \equiv \iint d\boldsymbol{\theta} d\boldsymbol{\theta}' I(\boldsymbol{\theta}) I(\boldsymbol{\theta}') \Phi\left(\frac{b(\boldsymbol{\theta}, \boldsymbol{\theta}')}{r_\perp}\right), \quad (38)$$

with limits $\Phi_{\text{eff}} \rightarrow 1$ (point-like source, or $r_\perp \gg D \Delta\theta$) and $\Phi_{\text{eff}} \rightarrow 0$ (fully resolved, $r_\perp \ll D \Delta\theta$).

B. Second-Order Coherence: Connected Wick Pairing

The unnormalized intensity correlator $\langle I(t) I(t+\tau) \rangle$ is a four-point function in E_{obs} . Conditioning on the realization of δt and applying Wick's theorem to the independent zero-mean complex-Gaussian $E_{\boldsymbol{\theta}}$ produces two pairings; spatial incoherence then collapses the four-fold angular integral to a double integral over ray pairs. The disconnected pairing ($\boldsymbol{\theta}_1 = \boldsymbol{\theta}_2, \boldsymbol{\theta}_3 = \boldsymbol{\theta}_4$) contributes $\langle I(t) \rangle \langle I(t+\tau) \rangle$ and cancels in $g^{(2)} - 1$. The connected pairing forces $\boldsymbol{\theta}_1 = \boldsymbol{\theta}_4 \equiv \boldsymbol{\theta}$ and $\boldsymbol{\theta}_3 = \boldsymbol{\theta}_2 \equiv \boldsymbol{\theta}'$, and yields two surviving two-point functions evaluated at retarded-time differences that depend on *two same-direction jitter increments*:

$$X \equiv \delta t(t + \tau; \boldsymbol{\theta}) - \delta t(t; \boldsymbol{\theta}), \quad Y \equiv \delta t(t + \tau; \boldsymbol{\theta}') - \delta t(t; \boldsymbol{\theta}'). \quad (39)$$

X is the temporal jitter increment along direction $\boldsymbol{\theta}$; Y is the analogous increment along direction $\boldsymbol{\theta}'$. Neither involves a cross-direction sample of δt ; the transverse coherence enters only through their joint distribution. Using $g_{\text{src}}^{(1)}(-u) = g_{\text{src}}^{(1)*}(u)$, the connected piece is

$$g_{\text{obs}}^{(2)}(\tau) - 1 = \iint d\boldsymbol{\theta} d\boldsymbol{\theta}' I(\boldsymbol{\theta}) I(\boldsymbol{\theta}') \langle g_{\text{src}}^{(1)}(\tau - X) g_{\text{src}}^{(1)*}(\tau - Y) \rangle_{X,Y}. \quad (40)$$

The full Wick contraction is carried out in Appendix A.

X and Y are linear functionals of the Gaussian process δt , hence jointly Gaussian and zero-mean, with

$$\text{Var}(X) = \text{Var}(Y) = \frac{2\sigma_\ell^2(D)}{c^2} [1 - C_\alpha(\tau; D)], \quad \text{Cov}(X, Y) = \frac{2\sigma_\ell^2(D)}{c^2} [1 - C_\alpha(\tau; D)] \Phi(b/r_\perp), \quad (41)$$

so the correlation coefficient is $\rho = \Phi(b/r_\perp)$.

Equation (40) is a 2D average over the joint distribution of (X, Y) , not a 1D convolution with a single jitter variable. It collapses to a 1D convolution only at $\Phi = 1$ (point source, or $b \ll r_\perp$): there $X = Y$ almost surely, $g_{\text{src}}^{(1)}(\tau - X) g_{\text{src}}^{(1)*}(\tau - X) = |g_{\text{src}}^{(1)}(\tau - X)|^2 = g_{\text{src}}^{(2)}(\tau - X) - 1$, and the point-source formula Eq. (24) is recovered with $\sigma_\Delta^2 = \text{Var}(X) = (2\sigma_\ell^2/c^2)[1 - C_\alpha(\tau)]$.

C. Finite-Lag Suppression

a. Zero lag. At $\tau = 0$ each of the increments in Eq. (39) is the difference of a random variable with itself: $X = 0$ and $Y = 0$ *identically*, for every realization of δt and every $(\boldsymbol{\theta}, \boldsymbol{\theta}')$ — independent of Φ , C_α , σ_ℓ , and D . Equation (40) reduces to $\iint I I' |g_{\text{src}}^{(1)}(0)|^2 = 1$, so

$$g_{\text{obs}}^{(2)}(0) = 2 \quad \text{exactly}, \quad (42)$$

and $\mathcal{S}(0) = g_{\text{obs}}^{(2)}(0) - 2 = 0$ for any extended source in this model. The conditional Gaussianity of $E_{\text{obs}}(t)$ given δt (see Appendix A) preserves the single-mode bunching peak exactly; foam imprints itself on multi-time correlations but not on the single-time amplitude statistics.

b. Finite lag. For $\tau \neq 0$ the joint (X, Y) average develops nontrivial structure. It is useful to split the jitter into the pair-average $U \equiv (X + Y)/2$, which displaces both envelope arguments by the same amount, and the half-difference $V \equiv (X - Y)/2$, which modulates only the relative carrier phase between the two ray directions. Splitting $g_{\text{src}}^{(1)}(\tau) = e^{i\omega_0\tau} V_{\text{src}}(\tau)$ into carrier and envelope and changing to these variables (independent Gaussians of variances $\sigma_U^2 = (\sigma_\ell^2/c^2)[1 - C_\alpha](1 + \Phi)$ and $\sigma_V^2 = (4\sigma_\ell^2/c^2)[1 - C_\alpha](1 - \Phi)$; see Appendix A), the narrow-band reduction yields

$$g_{\text{obs}}^{(2)}(\tau) - 1 = \iint d\boldsymbol{\theta} d\boldsymbol{\theta}' I(\boldsymbol{\theta}) I(\boldsymbol{\theta}') \mathcal{F}_{\text{cross}}(\tau; b) \int dU \mathcal{N}(U; 0, \sigma_U^2) V_{\text{src}}^2(\tau - U), \quad (43)$$

with the *cross-direction carrier-phase factor*

$$\mathcal{F}_{\text{cross}}(\tau; b) \equiv \exp\left(-2\sigma_\phi^2(D) [1 - C_\alpha(\tau; D)] [1 - \Phi(b/r_\perp)]\right), \quad (44)$$

which reduces to unity at $\tau = 0$ (recovering Eq. (42)), at $\Phi = 1$ (recovering the point-source formula Eq. (24)), and in the quasi-static foam limit $C_\alpha \rightarrow 1$. The decoupling between the imaging-channel scale σ_ϕ and the HBT scale σ_Δ/τ_c established in Section III F 2 is exact only for a point source: an extended source viewed at finite lag picks up σ_ϕ in $g^{(2)}$ through $\mathcal{F}_{\text{cross}}$.

c. Siegert discrepancy. Inserting Eqs. (43) and (37) into the Siegert-discrepancy definition $\mathcal{S}(\tau) \equiv [g_{\text{obs}}^{(2)}(\tau) - 1] - |g_{\text{obs}}^{(1)}(\tau)|^2$, expanding in the weak-foam regime $\sigma_\phi^2[1 - C_\alpha(\tau)] \ll 1$, and performing the angular integral gives

$$\mathcal{S}_{\text{ext}}(\tau) \approx \Phi_{\text{eff}} \cdot \mathcal{S}_{\text{pt}}(\tau), \quad (45)$$

where $\mathcal{S}_{\text{pt}}(\tau) \approx 2\sigma_\phi^2[1 - C_\alpha(\tau)] V_{\text{src}}^2(\tau)$ is the point-source Siegert discrepancy of Section III F. The extended-source Siegert discrepancy is therefore *suppressed* by $\Phi_{\text{eff}} \in [0, 1]$ relative to the point-source value: the point-source result is the upper bound on the foam signature in this framework. Targets believed to be intrinsically compact ($D\Delta\theta \ll r_\perp$ for any plausible $r_\perp$) saturate the upper bound and are the safest detection candidates; partially resolved targets degrade the SNR by the same factor Φ_{eff} , and the angular-size dependence is itself a positive discriminant — comparing $\mathcal{S}(\tau)$ across targets of varying angular size at fixed D probes $r_\perp$ through the profile $\Phi(b/r_\perp)$.

D. Discriminants Against Conventional Phase Screens

Before turning to sensitivity estimates, we note the model-independent signatures that distinguish foam from conventional propagation phase screens. The Siegert observable serves as such a discriminant [3]; for foam models with long-range transverse correlations — where image blurring is strongly suppressed [10] — it may be the only sensitive channel. The mathematical structure of the foam suppression factor $\mathcal{F}_{\text{foam}}$ is identical to that of any Gaussian random phase screen, but two signatures distinguish foam: distance scaling and achromaticity. For spacetime foam $\sigma_\phi^2 \propto D^{2(1-\alpha)}$ (D^1 for $\alpha = 1/2$, $D^{2/3}$ for $\alpha = 2/3$, D^0 for $\alpha = 1$), whereas a thin intervening screen scales as D and Kolmogorov turbulence as $D^{11/6}$; the holographic and conservative foam models predict sub-linear or flat growth qualitatively different from any known astrophysical screen. Geometrically, foam jitter ($\delta t = \Delta\ell/c$) is achromatic — independent of λ — in contrast to plasma-based effects which scale as λ^2 . The Fermi, H.E.S.S., and LHAASO time-of-flight analyses [20–22] rule out deterministic, energy-ordered LIV dispersion as a class of quantum-gravity effect but do not constrain the stochastic achromatic foam jitter considered here (Section I); the principal chromatic alternative the achromaticity discriminant addresses is therefore plasma propagation. In practice, the intrinsic source visibility $V_{\text{src}}(\tau)$ introduces a degeneracy with foam suppression, but $\mathcal{S}(\tau)$ is constructed directly from the data without a separate source-spectrum model (Section III F).

V. SIEGERT-VIOLATION: SENSITIVITY AND TARGET SELECTION

This section examines the detectability of Siegert violation via the HBT split-beam configuration [11, 23–25] and surveys candidate sources with Lorentzian (or Lorentzian-dominated) emission lines for which the sensitivity estimates apply directly. The single foam observable is the finite-lag Siegert discrepancy $\mathcal{S}(\tau)$ of Section III F; the point-source case of Section V B gives its maximum-signal limit, and the extended-source case of Section V C reports the same observable scaled by Φ_{eff} . We treat the resulting SNR figures as a statistical floor; in practice systematic errors will set the binding limit and a dedicated systematic budget is required for any claimed detection.

A. Sensitivity and Aperture Scaling

For a single-aperture HBT configuration using a 50/50 beamsplitter with unpolarized light [26], the per-lag noise on $g_{\text{obs}}^{(2)}$ is

$$\sigma[g_{\text{obs}}^{(2)}] = \frac{4}{R\sqrt{\tau_r T_{\text{obs}}}}, \quad (46)$$

where R is the total incident photon count rate, τ_r the timing resolution (correlation bin width) of the detectors, and T_{obs} the total observation time. The prefactor $4 = 2N_p$ with $N_p = 2$ polarization modes reflects that only photons sharing a polarization state contribute to the bunching signal, while the shot noise is set by the total (polarization-summed) count rate. The SNR for detecting a Siegert discrepancy of amplitude $\mathcal{S}(\tau)$ evaluated at a specific lag τ — treating $\mathcal{S}$ as constant within the bin of width τ_r — is then

$$\text{SNR}_{\mathcal{S}} = \frac{\mathcal{S}}{\sigma[g_{\text{obs}}^{(2)}]} = \frac{\mathcal{S} R}{4} \sqrt{\tau_r T_{\text{obs}}}. \quad (47)$$

All numerical projections adopt $A_\alpha = 1$ and the dynamical-foam limit $C_\alpha(\tau_L; D) \rightarrow 0$. The Siegert observable requires $\tau_{\text{foam}} \lesssim \tau_L$ so that $1 - C_\alpha(\tau_L)$ is of order unity; in the quasi-static limit $C_\alpha \rightarrow 1$ the signal vanishes for all targets and the channel becomes inoperative.

The photon-counting rate is $R = \epsilon A \dot{n}$, where $\dot{n}$ is the source photon flux per unit area within the filter bandpass, ϵ the end-to-end throughput, and A the collecting area. We adopt the Kim *et al.* [27] baseline: $d = 9.96$ m ($A = 77.9$ m²), $\epsilon = 0.39$, and $\sigma_t = 13$ ps rms (30 ps FWHM). The astrophysical targets of interest have intrinsic coherence times $\tau_L \lesssim 1$ ps, so $\sigma_t \gg \tau_L$ for the baseline instrument and the measurement sits firmly in the detector-limited regime: an unresolved foam feature of width τ_L in a bin of width σ_t contributes an effective signal $(\tau_L/\sigma_t) |S|$, so Eq. (47) becomes

$$\text{SNR}_{\text{det}} = \frac{|S| R \tau_L}{4} \sqrt{\frac{T_{\text{obs}}}{\sigma_t}}. \quad (48)$$

We present the detector-limited SNR as the primary case throughout and quote the source-limited limit ($\tau_L \gtrsim \sigma_t$, requiring sub-ps timing) only as the gain that next-generation detectors would unlock.

Achieving a statistically significant SNR is a necessary but not sufficient condition for a foam detection. A separate and equally demanding challenge is controlling systematic errors—spurious correlations from detector afterpulsing, electronic crosstalk, and optical back-reflections produce positive artifacts near zero lag, atmospheric scintillation introduces classical intensity fluctuations that mimic $g^{(2)} > 1$, and detector dead time causes a compensating dip in $g^{(2)}$ near zero lag—to levels below the expected foam signal $|S|$. Because the foam signatures are small ($|S| \ll 1$ for all viable foam models), even modest systematics can mimic or mask the effect.

B. Finite-Lag Channel: Point Sources

For a point-like Lorentzian source in the fast-foam / dynamical limit ($\tau_{\text{foam}} \ll \tau_L$, $C_\alpha \rightarrow 0$ above τ_{foam}), the Siegert discrepancy evaluated at its peak lag τ_{max} (Eq. (32)) is given by Eq. (33), $\mathcal{S}(\tau_{\text{max}}) \approx 1 - e^{-2\sigma_\phi^2}$ with σ_ϕ the rms accumulated phase (Eq. (22)): quadratic in σ_ϕ in the weak-foam limit ($\mathcal{S}(\tau_{\text{max}}) \approx 2\sigma_\phi^2$) and saturating at 1 for $\sigma_\phi \gg 1$.

a. Detector-limited SNR. Substituting $\mathcal{S}(\tau_{\text{max}})$ into Eq. (48),

$$\text{SNR}_{\text{det}}^{\text{pt}} = \frac{|\mathcal{S}(\tau_{\text{max}})| R \tau_L}{4} \sqrt{\frac{T_{\text{obs}}}{\sigma_t}} \xrightarrow{\sigma_\phi \ll 1} \frac{1}{2} \sigma_\phi^2 R \frac{\tau_L}{\sqrt{\sigma_t}} \sqrt{T_{\text{obs}}}. \quad (49)$$

The τ_L factor here is the source coherence width (the natural HBT coincidence-bin scale), not the evaluation lag. The detector-limited form carries a $\sqrt{\tau_L/\sigma_t}$ penalty relative to source-limited operation ($\text{SNR}_{\text{src}}^{\text{pt}} = \frac{1}{4} |\mathcal{S}(\tau_{\text{max}})| R \sqrt{\tau_L T_{\text{obs}}}$); reaching the source-limited limit on broad-line quasars requires $\sigma_t \lesssim \tau_L \sim 0.1$ ps.

b. Filter strategy. The foam signal $\mathcal{S}(\tau_{\text{max}})$ depends only on σ_ϕ , not on the filter width. The SNR (Eq. (49)) scales as $R \tau_L$, and these two quantities trade off against each other as the filter is narrowed: a narrower filter reduces the transmitted flux R but increases the post-filter coherence time $\tau_c = \lambda^2/(c \Delta \lambda_{\text{filter}})$. For a Lorentzian emission line, $R \propto \Delta \lambda_{\text{filter}}$ and $\tau_c \propto \Delta \lambda_{\text{filter}}^{-1}$ in the narrow-filter limit, so $R \tau_c$ is constant and the SNR is insensitive to filter width. The same cancellation holds for a flat continuum source. The practical lower bound on filter width is sky background: widening the filter beyond the emission line adds sky photons without increasing the source signal.

c. Distance scaling. Substituting $R \propto D^{-2}$ at fixed luminosity and $\sigma_\phi^2 \propto D^{2(1-\alpha)}$ into Eq. (49) gives, in the weak-foam limit,

$$\text{SNR}^{\text{pt}} \propto D^{2(1-\alpha)} \cdot D^{-2} = D^{-2\alpha}, \quad (50)$$

saturating at $\text{SNR} \propto D^{-2}$ in the strong-foam limit. Since $\alpha > 0$, the SNR increases toward smaller D : nearby targets are preferred, provided the flat-space ($z \ll 1$) limit of Eq. (22) holds. At cosmological distances the D -dependence is modified by two competing effects. For a fixed rest-frame line, $\lambda_{\text{obs}} = \lambda_{\text{rest}}(1+z)$ grows with redshift, reducing σ_ϕ via the $1/\lambda_{\text{obs}}^2$ prefactor in Eq. (22). Conversely, a fixed observer-frame bandpass accesses progressively bluer rest-frame transitions at higher redshift ($\lambda_{\text{rest}} = \lambda_{\text{obs}}/(1+z)$); since $\sigma_\phi \propto 1/\lambda_{\text{rest}}$, a source whose spectral line falls in the UV rest-frame accumulates substantially more phase than a lower-redshift counterpart observed at the same detector wavelength—potentially reversing the $D^{-2\alpha}$ preference. Resolving this competition requires a target-specific spectral-energy-distribution model; the entries in Table I are evaluated at the observed wavelength of each emission line, which should be treated as accurate to a factor of a few in σ_ϕ for sources at $z \gtrsim 1$.

The minimum detectable path-length amplitude follows from inverting Eq. (49) with the quadratic signal dependence $|\mathcal{S}(\tau_{\max})| \propto \sigma_\phi^2 \propto \sigma_\ell^2$:

$$\sigma_{\ell, \min}^{\text{pt}} \propto \frac{\lambda}{d} \left(\frac{\sigma_t}{\tau_L^2 T_{\text{obs}}} \right)^{1/4}, \quad (51)$$

where $R \propto d^2$ has been used. Sensitivity improves as d^{-1} rather than d^{-2} , and as $T_{\text{obs}}^{-1/4}$ rather than $T_{\text{obs}}^{-1/2}$ — both costs of the quadratic dependence on σ_ϕ . Each additional decade of sensitivity in σ_ℓ requires four decades of integration time, compared with two for the extended-source variant.

C. Finite-Lag Channel: Extended Sources

For an extended source viewed at finite lag, the Siegert discrepancy is suppressed relative to the point-source value by the angular-size factor Φ_{eff} of Eq. (38), which weights the transverse coherence function $\Phi(b/r_\perp)$ over ray pairs: directions separated by more than $r_\perp$ contribute jitter increments X and Y (Section IV B) that are nearly uncorrelated, broadening the effective convolution kernel without adding Siegert signal. In the weak-foam regime relevant to all viable foam models, Eq. (45) gives

$$\mathcal{S}_{\text{ext}}(\tau_{\max}) \approx \Phi_{\text{eff}} \cdot \mathcal{S}_{\text{pt}}(\tau_{\max}) \approx 2 \Phi_{\text{eff}} \sigma_\phi^2, \quad (52)$$

and the detector-limited SNR follows by scaling Eq. (49):

$$\text{SNR}_{\text{det}}^{\text{ext}} = \Phi_{\text{eff}} \cdot \text{SNR}_{\text{det}}^{\text{pt}}. \quad (53)$$

The point-source SNR of Section V B is the *maximum-signal upper bound* for the foam observable in this framework; no target gives SNR above the point-source value. The extended-source factor $\Phi_{\text{eff}} \in [0, 1]$ scales this upper bound down for partially resolved targets and depends on the unknown foam transverse coherence length $r_\perp$: $\Phi_{\text{eff}} \rightarrow 1$ when $D\Delta\theta \ll r_\perp$ (the source is unresolved by the foam, recovering the upper bound), and $\Phi_{\text{eff}} \rightarrow 0$ when $D\Delta\theta \gg r_\perp$ (the source is fully resolved, signal vanishes).

a. Caveat on $r_\perp$. Because $r_\perp$ is an unknown foam-model parameter, the extended-source SNR is reported as a band over plausible $r_\perp$ rather than as a single number. Targets believed to be intrinsically compact (e.g., narrow-line AGN cores with $D\Delta\theta$ smaller than any plausible $r_\perp$) saturate the point-source bound and provide the most conservative detection candidates; partially resolved targets degrade the SNR by Φ_{eff} , and a non-detection in such a target is consistent either with foam below threshold or with $r_\perp$ small enough to resolve the source.

b. Angular-size discriminant. Should a signal be detected, the source-size dependence of $\mathcal{S}(\tau)$ across targets of varying $D\Delta\theta$ is a positive observational handle on $r_\perp$, with no analogue in a featureless propagation screen (Section IV). A multi-target campaign probes $r_\perp$ through the profile $\Phi(b/r_\perp)$ even when no individual measurement does.

c. Distance scaling. The distance scaling of Eq. (50) is unchanged for extended targets up to the Φ_{eff} prefactor: $\text{SNR}_{\text{det}}^{\text{ext}} \propto \Phi_{\text{eff}} \cdot D^{-2\alpha}$ in the weak-foam limit. Filter strategy, τ_L scaling, and the minimum detectable path-length amplitude all carry over directly from Section V B with an overall Φ_{eff} factor in the signal.

D. Candidate Sources

Two source properties drive target selection: the photon rate R , set by the total line flux captured within the filter, and the source distance D , which sets σ_ϕ via Eq. (22) and hence the signal amplitude $\mathcal{S}(\tau_{\max}) \approx 1 - e^{-2\sigma_\phi^2}$ of Eq. (33). For partially resolved targets the signal is further suppressed by Φ_{eff} (Section V C), which depends on the unknown foam transverse coherence length $r_\perp$; the entries below quote the point-source upper bound and flag Φ_{eff} caveats where the source has substantial angular extent. All sources discussed below have intrinsic coherence times $\tau_L \lesssim 1$ ps, well below the baseline timing resolution $\sigma_t = 13$ ps; the measurements therefore operate in the detector-limited regime throughout. The source classes surveyed below are illustrative rather than exhaustive; systematic searches through existing AGN flux catalogues and emission-line surveys will likely identify superior targets with higher photon rates, more purely Lorentzian line profiles, or more favorable distance-to-flux ratios. Observed properties and references for the principal candidates are collected in Table I.

Narrow-line Seyfert 1 galaxies as headline AGN targets. NLS1s (Population A; FWHM $\lesssim 4000$ km s $^{-1}$; [37, 38]) have single-Lorentzian broad lines and intrinsically compact broad-line regions, both favourable for the Siegert observable.

TABLE I. Candidate sources for the sensitivity analysis of Section V. D is the luminosity distance (flat Λ CDM, Planck 2018 [28]); m is the apparent magnitude (band per source class); F is the integrated line flux; FWHM is observed-frame; $\tau_L = \lambda_{\text{obs}}^2 / (\pi c \Delta \lambda_{\text{FWHM}})$. Per-source caveats (sample-mean SN IIn scaling, η Car factor-of-a-few uncertainty, J1330–0905 image-A flux derivation) are given in Section VD. [†]Adopted from the Kollatschny & Zetzl 2013 turbulent FWHM [29]; not resolved by the XP spectrum.

Source	z	D (Mpc)	m	Line	F (erg s ^{−1} cm ^{−2})	FWHM (Å)	τ_L (ps)
Mrk 766	0.013	59	13.6	H α	2.8×10^{-12} [30]	28 [29]	0.17
PHL 1811	0.192	971	14.4	H α	1.0×10^{-12} [31]	46 [31]	0.14
η Car (1843)	0	2.3×10^{-3} [32]	−1	H α	$\sim 5 \times 10^{-7}$ [33]	15 [33]	0.30
Gal. SN IIn	0	2×10^{-3}	−7.6	H α	$\sim 2 \times 10^{-4}$ [34, 35]	55 [34]	0.083
M31 SN IIn	0	0.78	5.3	H α	$\sim 1.4 \times 10^{-9}$ [34, 35]	55 [34]	0.083
J1330–0905 (A)	2.225	1.81×10^4	16.1	Ly α	5.1×10^{-13} [36]	52 [†]	0.031

The NLS1 BLR has angular size $\sim 10^{-6}$ arcsec at cosmological distances, far below any plausible $r_{\perp}$, so $\Phi_{\text{eff}} \approx 1$ and these targets saturate the point-source upper bound. We adopt two reference targets at complementary distances — the nearby Mrk 766 and the cosmological PHL 1811 (Table I; luminosity distances in flat Λ CDM with the Planck 2018 cosmology [28]); the H α and H β profiles of PHL 1811 are explicitly Lorentzian-fit by Leighly et al. 2007 (their §3) [31]. A wide filter ($\Delta \lambda_{\text{filter}} \geq \text{FWHM}$) captures the full line flux, giving photon rates $R \approx 2.0 \times 10^5$ cps (Mrk 766) and $R \approx 8.5 \times 10^4$ cps (PHL 1811) on a 10 m telescope with $\epsilon = 0.39$; sky background and detector dark counts are negligible relative to the source rates, so the Brouni et al. correction for accidental coincidences is small [39]. For both targets the $2 \times \text{FWHM}_{\text{line}}$ filter width is narrow enough that the underlying pseudo-continuum (Fe II forest plus near-line continuum) dilutes the signal by $\lesssim 10\%$ given the line-to-continuum ratio $\gtrsim 10:1$ typical of these objects; the dilution does not change the qualitative SNR ordering but should be subtracted from the measured $g^{(2)}(\tau)$ as an additive constant when extracting $\mathcal{S}(\tau)$ in practice. The two targets span a factor of 16 in distance, exercising the $\sigma_{\phi} \propto D^{1-\alpha}$ scaling of the foam-vs-screen discriminant developed in Section IV.

Strongly-lensed quasars. Gravitational magnification by factors of ~ 10 –60 boosts the photon rate while preserving the cosmological foam-variance lever arm at the source’s intrinsic distance, making strongly-lensed quasars a complementary persistent-AGN target class. Persephone’s Torch (J1330–0905, $z = 2.22$, total magnification ~ 56 , brightest-image $i \approx 16$) [36] is included in Table I on the basis of its broad Ly α emission line measured directly from its GAIA DR3 XP spectrum (observed equivalent width 235 Å, integrated 3700–4100 Å line flux 1.28×10^{-12} erg s^{−1} cm^{−2} for the GAIA blob; image A carries a fraction 0.40 of the total system flux from the observed ratios $F_B/F_A = 0.74$, $F_C/F_A = 0.70$, $F_D/F_A = 0.08$, giving 5.1×10^{-13} erg s^{−1} cm^{−2} on image A). Under the Kollatschny & Zetzl 2013 [29] broad-line-region model the observed Ly α profile is a Voigt profile with a Lorentzian core from BLR turbulent broadening ($\text{FWHM}_{\text{turb}} \sim 3800$ km s^{−1}) convolved with a rotational Gaussian; the Lorentzian core sets the coherence time τ_L relevant for the finite-lag Siegert observable. The BLR angular size at $z = 2.2$ is $\sim 10^{-7}$ arcsec, far below any plausible $r_{\perp}$, so $\Phi_{\text{eff}} \approx 1$ and this target saturates the point-source upper bound. The net finite-lag SNR exceeds PHL 1811 by $\sim 1.4\times$ at $\alpha = 2/3$ (Figure 1). The Lorentzian-core hypothesis at high z is an extrapolation from the low- z K&Z sample; direct high-resolution spectroscopy of Ly α (or the redshifted near-infrared Balmer lines) would confirm it.

Galactic and Local-Group transients. Type IIn supernovae and LBV giant eruptions — with η Car as the principal example — combine high H α luminosity with narrow Lorentzian line cores ($\tau_L \sim 0.1$ –0.3 ps). The point-source upper-bound SNR scales as $\sigma_{\phi}^2 \propto D^{2(1-\alpha)}$ and is small at kpc–Mpc distances; however, the giant photon rate R at close range partially compensates, and the resulting SNR remains competitive for $\alpha \rightarrow 1$ (conservative foam). At smaller α these nearby targets lose the cosmological lever arm and fall below the AGN baselines (Section VE).

a. Caveats for transient targets. Three effects can degrade the actual SNR below the plotted upper bounds. (i) *Source extent*: ejecta-photosphere angular sizes of $\sim 10^{-5}$ – 10^{-3} arcsec can exceed $r_{\perp}$ for small foam-coherence lengths, suppressing the signal by $\Phi_{\text{eff}}(r_{\perp})$; this is the most constraining caveat for nearby events. (ii) *Scattering geometry*: the Homunculus Nebula around η Car scatters substantial flux into the aperture; the scattering geometry must be modeled before crediting the full photon count toward the Siegert observable. (iii) *Stationarity*: the 10 hr per-event integrations used in the figures satisfy the assumption of a stationary Gaussian source, but 10^4 hr campaign-scale totals require co-adding over stationary epochs (Section VE).

Type IIn supernovae. Rather than a specific event, we adopt a representative Type IIn source characterized by the sample-mean peak H α luminosity $L(\text{H}\alpha) = 10^{41}$ erg s^{−1} [34], broad-line FWHM ≈ 55 Å (H α rest-frame) [34], and apparent r -band magnitude from the sample-mean absolute magnitude $M_r = -19.16$ [35]. The H α flux at a given distance follows by inverse-square scaling; we evaluate two cases: a Galactic event at $D = 2$ kpc and an M31 event at $D = 778$ kpc (Table I).

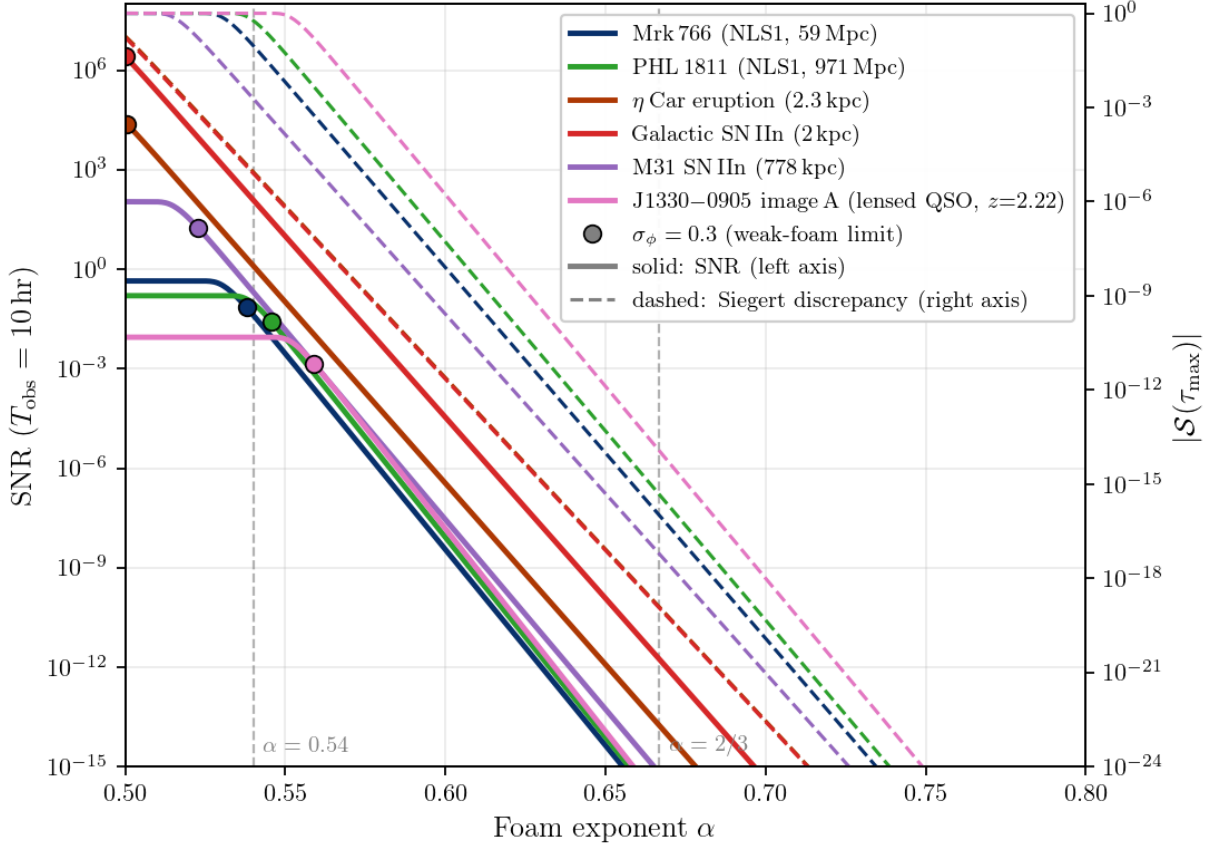


FIG. 1. Maximum-signal (point-source) finite-lag Siegert discrepancy SNR for $\mathcal{S}(\tau_{\max}) \approx 1 - e^{-2\sigma_\phi^2}$ (Eq. (33)) as a function of foam exponent α , in the fast-foam / dynamical limit $C_\alpha \rightarrow 0$. The values shown are the upper bound for each target; partially resolved targets ($D\Delta\theta \gtrsim r_\perp$) are suppressed by $\Phi_{\text{eff}} \in [0, 1]$ (Section V C). Kim et al. [27] baseline (10 m, $\epsilon = 0.39$, $\sigma_t = 13$ ps rms), $T_{\text{obs}} = 10$ hr per exposure (multiply SNR by $\sqrt{10^3} \approx 31.6$ for the 10^4 hr campaign quoted in Sections V E and VI); detector-limited regime (Eq. (49)). Filled circles mark the $\sigma_\phi = 0.3$ weak-foam validity boundary; curves to the right saturate at 1. Vertical dashed lines: $\alpha = 0.54$ (Lieu–Hillman imaging exclusion at $D = 1$ Gpc, $\lambda = 500$ nm) and $\alpha = 2/3$ (holographic). J1330–0905 is evaluated on the GAIA DR3 XP-measured Ly α flux with the Kollatschny & Zetzl 2013 turbulent-Lorentzian core (Section V D). Dashed curves (right axis): $|\mathcal{S}(\tau_{\max})|$.

E. SNR Results

Figure 1 shows the maximum-signal SNR for all six targets in Table I; the main features of the distribution are the following.

a. Strong-foam regime ($\alpha \lesssim 0.54$). At and below the Lieu–Hillman imaging exclusion the signal saturates at $\mathcal{S}(\tau_{\max}) \rightarrow 1$ (Eq. (33)), producing the flattening of every SNR curve visible in Fig. 1 at small α . On the plateau the SNR is set entirely by the photon rate R and is independent of α . For the highest- R targets the saturation level itself is detectable in $T_{\text{obs}} = 10$ hr: a Galactic SN IIn at $\alpha = 0.54$ reaches SNR ~ 110 , and an η Car-class eruption is detectable at SNR ~ 1.2 (each modulo Φ_{eff} for substantial angular extent). The HBT framework is therefore statistically competitive with — and methodologically complementary to — the imaging tests that set the $\alpha = 0.54$ boundary, with the advantage of sensitivity to foam models with long-range transverse correlations that evade those tests [10].

b. Holographic regime ($\alpha = 2/3$). All signals and SNRs lie many orders of magnitude below unity. On Mrk 766, $\sigma_\phi \sim 7.3 \times 10^{-9}$ gives $\mathcal{S}(\tau_{\max}) \sim 1.1 \times 10^{-16}$, and the SNR at $T_{\text{obs}} = 10$ hr is $\sim 4.9 \times 10^{-17}$. The finite-lag channel is therefore the only HBT probe in this regime and is not competitive with the imaging upper bounds on its own; gains come from photon-rate-leading targets (high- R transients) or from ultra-long campaigns combined with sub-ps timing.

c. Persistent AGN: Mrk 766 and PHL 1811. The two NLS1 targets span a factor of 16 in distance and trade off photon rate against cosmological lever arm. PHL 1811’s $D^{1-\alpha}$ boost in σ_ℓ at $\alpha = 2/3$ approximately compensates its D^{-2} flux deficit, leaving the two within $\sim 10\%$ of each other (Mrk 766 nominally higher; see Fig. 1). Their near-degeneracy makes the pair a useful internal consistency check on the $\sigma_\phi \propto D^{1-\alpha}$ scaling of the foam-vs-screen discriminant (Section IV). Both are intrinsically compact ($\Phi_{\text{eff}} \approx 1$) and saturate the upper bound.

d. Lensed quasar J1330–0905: largest $|\mathcal{S}|$ of the persistent targets. The strongly-lensed quasar carries the cosmological lever arm to $z = 2.22$, giving $\sigma_\phi \approx 7.8 \times 10^{-8}$ on image A at $\alpha = 2/3$ — roughly $5\times$ that of PHL 1811. The resulting signal $|\mathcal{S}(\tau_{\text{max}})| \sim 1.2 \times 10^{-14}$ is $\sim 25\times$ that of PHL 1811 and the largest of the six sources, despite the lower photon rate from a single magnified image (dashed curves, right axis of Fig. 1). As with the nearby NLS1s, the BLR is intrinsically compact and saturates the upper bound. For any HBT campaign that can absorb the photon-rate cost, J1330–0905 sets the high-water mark for foam signal amplitude among persistent targets.

e. Galactic and Local-Group transients. For a Galactic transient at $D \sim 2$ kpc, the modest cosmological lever arm makes the upper-bound signal small in absolute terms, but the giant photon rate R can compensate. The Galactic SN IIn case yields the largest single-event upper-bound SNR ($\sim 2 \times 10^6$ times the Mrk 766 baseline at $\alpha = 2/3$); the recurrence rate makes it a “drop-everything” opportunity rather than a campaign target. Smith et al. 2011 [40] estimate that $\sim 9\%$ of core-collapse supernovae are Type IIn, and combined with the Galactic core-collapse rate of 1.6 ± 0.5 per century from Adams et al. 2013 [41] this implies a Galactic IIn rate of roughly one per 500–1500 yr; the most recent confirmed Galactic core-collapse supernova was Cassiopeia A (~ 1680 , Type IIb). The η Car case sacrifices a factor of ~ 10 in SNR but offers a recurrence plausibly within a working career. The M31 SN IIn is intermediate (plotted adjacent to its Galactic counterpart in Fig. 1 for direct comparison) and degrades the Galactic SNR by the inverse-square-distance factor $(778/2)^2 \approx 1.5 \times 10^5$. All three transients have ejecta-photosphere angular sizes that may exceed $r_\perp$ for small- $r_\perp$ foam models, in which case the actual SNR is suppressed by $\Phi_{\text{eff}} < 1$ below the upper bounds plotted; a non-detection in these targets does not by itself rule out foam, since it is degenerate with small $r_\perp$ (Section V C).

F. Sources Investigated but Not Competitive

The $\sigma_\phi \propto D^{1-\alpha}$ scaling means that any nearby source suffers a phase-variance penalty no realistic photon-rate advantage can recover: at $D = 1$ AU, σ_ϕ^2 is $\sim 10^{16}$ smaller than at $D = 1$ Gpc for $\alpha = 1/2$ and utterly negligible for $\alpha = 2/3$. *The Sun* illustrates the ceiling: a $1 \text{ \AA}$ filter on a 10 m telescope yields $r \sim 10^8$ cps but $\text{SNR}/\sqrt{T/\text{hr}} \sim 1.5 \times 10^{-3}$, and even $N \sim 3000$ -channel spectrograph multiplexing reaches only $\sim 7 \times 10^{-2}$ at $T_{\text{obs}} = 10^4$ hr. *Sirius B* (Stark-broadened H α , $D = 2.64$ pc) gives $\text{SNR}/\sqrt{T/\text{hr}} \approx 1.4 \times 10^{-2}$ at $\alpha = 1/2$, negligible at $\alpha = 2/3$; its main value is as a null-test calibrator, since a nearby DA white dwarf has a Lorentzian profile with negligible predicted foam signal for any plausible α .

VI. CONCLUSIONS

A single second-order coherence observable of spacetime foam emerges from this framework: the finite-lag Siegert discrepancy $\mathcal{S}(\tau) = g^{(2)}(\tau) - 1 - |g^{(1)}(\tau)|^2$, present when foam fluctuates on timescales shorter than the optical coherence time ($\tau_{\text{foam}} \lesssim \tau_L$). No foam-induced zero-lag bunching suppression arises in this model: at $\tau = 0$ the same-direction jitter increments vanish identically and $g_{\text{obs}}^{(2)}(0) = 2$ for any extended source viewed by a single-mode detector (Section IV; Appendix A). Its value lies not in raw sensitivity — the signal is at $\sim 10^{-16}$ for the holographic $\alpha = 2/3$ model, well below current reach — but in three structural advantages over existing fringe-visibility and Strehl-ratio tests: (i) $\mathcal{S}(\tau)$ is constructed directly from the data, with $g_{\text{src}}^{(1)}(\tau)$ obtained from the source spectrum via Wiener–Khinchin, so a detection requires no separate source model; (ii) the angular-size suppression factor $\Phi_{\text{eff}}(r_\perp) \in [0, 1]$ (Section IV) makes extended targets weaker than compact ones but converts source-size variability across the target ensemble into a probe of the foam transverse coherence length $r_\perp$, a parameter inaccessible to single-target measurements; (iii) the framework remains sensitive to foam models with long-range transverse correlations that evade imaging tests entirely [10], and supplies a chromatic discriminant against plasma scattering through the $\sigma_\phi \propto 1/\lambda$ scaling.

The framework identifies high-priority targets: intrinsically compact emitters that saturate the upper bound regardless of $r_\perp$ (narrow-line Seyfert 1 cores, lensed-quasar BLRs), and high-flux transients (Type IIn supernovae, LBV giant eruptions) whose photon-rate advantage compensates for the modest cosmological lever arm at kpc–Mpc distances. A standing rapid-response HBT plan — covering η Car, any galactic core-collapse supernova, and any nearby ($D \lesssim 30$ Mpc) Type IIn identified by transient surveys — would let the framework exploit such high-yield events

without prior allocation; transient targets with substantial angular extent contribute to the $r_\perp$ discriminant via Φ_{eff} . Strongly-lensed quasars (Section VD; J1330–0905, comparable to the NLS1 baseline after cosmological correction) round out the persistent target class.

The achievable sensitivities are derived in Section V. The single finite-lag channel applies to point-like and extended targets alike, the latter scaled by Φ_{eff} . What HBT-on-AGN cannot reach is the holographic prediction $\sigma_\phi \sim 10^{-8}$, many orders of magnitude below the most aggressive HBT projection and the natural target of tabletop holographic-noise interferometers [42]. Realising these projections demands parallel investment in systematic-error control: null-test calibration sources with negligible predicted foam signal are essential to verify that instrumental artifacts do not limit sensitivity before photon statistics do.

This analysis has focused on single-telescope HBT observations, which constrain foam through temporal intensity statistics along a single line of sight. Intensity interferometry with spatially separated baselines extends this by adding cross-baseline temporal and spatial correlations simultaneously. Such measurements could test the factorization ansatz of Section II — the assumed separability of longitudinal variance, temporal decorrelation, and transverse coherence — and would distinguish foam models in which these three scales are entangled from those in which they are not.

ACKNOWLEDGMENTS

This work was supported in part by the Director, Office of Science, Office of High Energy Physics of the U.S. Department of Energy under Contract No. DE-AC02-05CH11231.

The authors used Claude (Anthropic) and Valency Systems to assist with drafting and editing portions of this manuscript along with detailed reference searches. All content was reviewed and verified by the authors, who take full responsibility for the work.

We thank Robin Kaiser for helpful comments.

Appendix A: Pedestrian derivation of $g_{\text{obs}}^{(2)}(\tau)$

This appendix carries out the four-point function calculation underlying Section IV step by step. The aim is to expose how the random variables (X, Y) that enter the connected Wick pairing arise, and to identify the regime in which the 2D average of Eq. (40) collapses to a 1D convolution (the point-source limit $\Phi = 1$) versus the regime in which the cross-direction carrier-phase factor $\mathcal{F}_{\text{cross}}$ of Eq. (44) appears.

1. Setup

The observed field is Eq. (36),

$$E_{\text{obs}}(t) = \int d\boldsymbol{\theta} I^{1/2}(\boldsymbol{\theta}) E_{\boldsymbol{\theta}}(t - \delta t(t; \boldsymbol{\theta})), \quad (\text{A1})$$

with three statistical inputs:

1. Each $E_{\boldsymbol{\theta}}$ is a stationary, zero-mean, complex-Gaussian (analytic-signal) process with unit mean intensity $\langle |E_{\boldsymbol{\theta}}(s)|^2 \rangle = 1$.
2. Source spatial incoherence: $\langle E_{\boldsymbol{\theta}}^*(s) E_{\boldsymbol{\theta}'}(s') \rangle = g_{\text{src}}^{(1)}(s' - s) \delta(\boldsymbol{\theta} - \boldsymbol{\theta}')$, with anomalous pairings $\langle EE \rangle$ and $\langle E^* E^* \rangle$ vanishing.
3. δt is independent of E , and its joint distribution at distinct $(t, \boldsymbol{\theta})$ is the Gaussian process whose covariance is Eq. (4) divided by c^2 .

The brightness distribution is normalized, $\int I(\boldsymbol{\theta}) d\boldsymbol{\theta} = 1$, so $\langle I_{\text{obs}} \rangle = 1$ throughout this appendix.

2. Mean intensity

Plugging Eq. (A1) into $\langle |E_{\text{obs}}(t)|^2 \rangle$ and using spatial incoherence to collapse the angular integral to the diagonal,

$$\begin{aligned} \langle |E_{\text{obs}}(t)|^2 \rangle &= \iint d\boldsymbol{\theta} d\boldsymbol{\theta}' I^{1/2}(\boldsymbol{\theta}) I^{1/2}(\boldsymbol{\theta}') \langle E_{\boldsymbol{\theta}}^*(t - \delta t_1) E_{\boldsymbol{\theta}'}(t - \delta t_2) \rangle \\ &= \int d\boldsymbol{\theta} I(\boldsymbol{\theta}) \langle |E_{\boldsymbol{\theta}}(t - \delta t(t; \boldsymbol{\theta}))|^2 \rangle = \int d\boldsymbol{\theta} I(\boldsymbol{\theta}) = 1, \end{aligned} \quad (\text{A2})$$

where $\delta t_1 = \delta t(t; \boldsymbol{\theta})$ and $\delta t_2 = \delta t(t; \boldsymbol{\theta}')$. By stationarity, the random retarded time does not change the marginal second moment of each $E_{\boldsymbol{\theta}}$, so the foam preserves the mean intensity exactly: $\langle |E_{\text{obs}}|^2 \rangle = \langle |E_{\text{src}}|^2 \rangle$.

3. Four-point function and Wick contraction

The intensity autocorrelation is $\langle I(t) I(t + \tau) \rangle = \langle E_{\text{obs}}^*(t) E_{\text{obs}}(t) E_{\text{obs}}^*(t + \tau) E_{\text{obs}}(t + \tau) \rangle$. Expanding each E_{obs} as in Eq. (A1) introduces four direction labels $\boldsymbol{\theta}_1, \dots, \boldsymbol{\theta}_4$ and four retarded times

$$\begin{aligned} s_1 &= t - \delta t(t; \boldsymbol{\theta}_1), & s_2 &= t - \delta t(t; \boldsymbol{\theta}_2), \\ s_3 &= t + \tau - \delta t(t + \tau; \boldsymbol{\theta}_3), & s_4 &= t + \tau - \delta t(t + \tau; \boldsymbol{\theta}_4). \end{aligned} \quad (\text{A3})$$

Conditioning on a realization of δt , the four-point function of the independent zero-mean complex-Gaussian $E_{\boldsymbol{\theta}}$ obeys Wick's (Isserlis') theorem:

$$\langle E_{\boldsymbol{\theta}_1}^*(s_1) E_{\boldsymbol{\theta}_2}(s_2) E_{\boldsymbol{\theta}_3}^*(s_3) E_{\boldsymbol{\theta}_4}(s_4) \rangle_E = \langle E_1^* E_2 \rangle \langle E_3^* E_4 \rangle + \langle E_1^* E_4 \rangle \langle E_3^* E_2 \rangle. \quad (\text{A4})$$

a. Disconnected pairing. The first term in Eq. (A4) forces $\boldsymbol{\theta}_1 = \boldsymbol{\theta}_2$ and $\boldsymbol{\theta}_3 = \boldsymbol{\theta}_4$ via the spatial-incoherence delta. With $\boldsymbol{\theta}_1 = \boldsymbol{\theta}_2$ the retarded times coincide ($s_1 = s_2$), so the two-point function is $g_{\text{src}}^{(1)}(0) = 1$; the same holds for the $\boldsymbol{\theta}_3, \boldsymbol{\theta}_4$ pair. The contribution is $\int I(\boldsymbol{\theta}_1) d\boldsymbol{\theta}_1 \int I(\boldsymbol{\theta}_3) d\boldsymbol{\theta}_3 = 1 = \langle I(t) \rangle \langle I(t + \tau) \rangle$, which subtracts cleanly when forming $g^{(2)} - 1$.

b. Connected pairing. The second term forces $\boldsymbol{\theta}_1 = \boldsymbol{\theta}_4 \equiv \boldsymbol{\theta}$ and $\boldsymbol{\theta}_3 = \boldsymbol{\theta}_2 \equiv \boldsymbol{\theta}'$. The two surviving two-point functions are evaluated at retarded-time differences

$$s_4 - s_1 = \tau - X, \quad s_2 - s_3 = -(\tau - Y), \quad (\text{A5})$$

with the two *same-direction temporal increments*

$$X \equiv \delta t(t + \tau; \boldsymbol{\theta}) - \delta t(t; \boldsymbol{\theta}), \quad Y \equiv \delta t(t + \tau; \boldsymbol{\theta}') - \delta t(t; \boldsymbol{\theta}') \quad (\text{A6})$$

[Eq. (39) of the body]. Using $g_{\text{src}}^{(1)}(-u) = g_{\text{src}}^{(1)*}(u)$, the connected contribution to $g_{\text{obs}}^{(2)} - 1$ is

$$g_{\text{obs}}^{(2)}(\tau) - 1 = \iint d\boldsymbol{\theta} d\boldsymbol{\theta}' I(\boldsymbol{\theta}) I(\boldsymbol{\theta}') \langle g_{\text{src}}^{(1)}(\tau - X) g_{\text{src}}^{(1)*}(\tau - Y) \rangle_{X,Y}.$$

This is Eq. (40) of the body.

4. Joint distribution of (X, Y)

X and Y are linear functionals of the Gaussian process δt , hence jointly Gaussian and zero-mean. Using Eq. (4) and C_{α} even in τ ,

$$\text{Var}(X) = \text{Var}(Y) = \frac{2\sigma_{\ell}^2(D)}{c^2} [1 - C_{\alpha}(\tau; D)], \quad (\text{A7})$$

$$\text{Cov}(X, Y) = \frac{2\sigma_{\ell}^2(D)}{c^2} [1 - C_{\alpha}(\tau; D)] \Phi(b/r_{\perp}), \quad (\text{A8})$$

so the correlation coefficient is $\rho = \Phi(b/r_{\perp})$.

The structural point: X is the temporal jitter increment *along direction* $\boldsymbol{\theta}$, and Y is the increment *along direction* $\boldsymbol{\theta}'$. Neither involves a cross-direction sample of δt ; the transverse coherence factor Φ enters only through their correlation. Equation (40) is therefore a 2D average over the joint distribution of (X, Y) , not a 1D convolution with a single variable Δ .

It collapses to a 1D convolution only when $\rho = 1$, i.e. when $\Phi = 1$ (point source, or the transversely-coherent limit $b \ll r_\perp$). In that case $X = Y$ almost surely, $g_{\text{src}}^{(1)}(\tau - X)g_{\text{src}}^{(1)*}(\tau - X) = |g_{\text{src}}^{(1)}(\tau - X)|^2 = g_{\text{src}}^{(2)}(\tau - X) - 1$, and the point-source formula Eq. (24) is recovered with $\sigma_\Delta^2 = \text{Var}(X) = (2\sigma_\ell^2/c^2)[1 - C_\alpha(\tau)]$.

5. Zero lag: $g_{\text{obs}}^{(2)}(0) = 2$ exactly

At $\tau = 0$ each of the increments in Eq. (39) is the difference of a random variable with itself:

$$X|_{\tau=0} = 0, \quad Y|_{\tau=0} = 0, \quad (\text{A9})$$

for every realization of δt and every pair $(\boldsymbol{\theta}, \boldsymbol{\theta}')$. This is independent of Φ , C_α , σ_ℓ , and D . Equation (40) reduces to

$$g_{\text{obs}}^{(2)}(0) - 1 = \iint d\boldsymbol{\theta} d\boldsymbol{\theta}' I(\boldsymbol{\theta})I(\boldsymbol{\theta}') |g_{\text{src}}^{(1)}(0)|^2 = 1, \quad (\text{A10})$$

so $g_{\text{obs}}^{(2)}(0) = 2$ exactly.

The same conclusion follows from a direct argument that bypasses the Wick contraction. Conditioning on δt , $E_{\text{obs}}(t) = \int d\boldsymbol{\theta} I^{1/2}(\boldsymbol{\theta}) \tilde{E}_{\boldsymbol{\theta}}$ with $\tilde{E}_{\boldsymbol{\theta}} \equiv E_{\boldsymbol{\theta}}(t - \delta t(t; \boldsymbol{\theta}))$. By stationarity each $\tilde{E}_{\boldsymbol{\theta}}$ has the same distribution as $E_{\boldsymbol{\theta}}(t)$; since the $E_{\boldsymbol{\theta}}$ are independent across $\boldsymbol{\theta}$, so are the $\tilde{E}_{\boldsymbol{\theta}}$. Hence $E_{\text{obs}}(t)$ is a linear combination of independent zero-mean complex Gaussians with fixed weights $I^{1/2}(\boldsymbol{\theta})$, and is itself a zero-mean complex Gaussian. The Gaussian-moment identity gives

$$\langle |E_{\text{obs}}(t)|^4 | \delta t \rangle = 2 \langle |E_{\text{obs}}(t)|^2 | \delta t \rangle^2 = 2 \quad (\text{A11})$$

for every δt , so $g_{\text{obs}}^{(2)}(0) = \langle I^2 \rangle / \langle I \rangle^2 = 2$ unconditionally. Foam imprints itself on multi-time correlations of E_{obs} but not on its single-time amplitude statistics; the standard Gaussian identity $g^{(2)}(0) = 2$ for any zero-mean complex Gaussian field is preserved.

6. Finite lag: narrow-band reduction

For $\tau \neq 0$, split $g_{\text{src}}^{(1)}$ into carrier and envelope (Section III D), $g_{\text{src}}^{(1)}(\tau) = e^{i\omega_0\tau} V_{\text{src}}(\tau)$ with V_{src} real and slowly varying on the carrier scale, so

$$g_{\text{src}}^{(1)}(\tau - X) g_{\text{src}}^{(1)*}(\tau - Y) = e^{-i\omega_0(X-Y)} V_{\text{src}}(\tau - X) V_{\text{src}}(\tau - Y). \quad (\text{A12})$$

Change variables in the joint Gaussian average to the sum and difference

$$U \equiv \frac{1}{2}(X + Y), \quad V \equiv X - Y, \quad (\text{A13})$$

which (because X, Y have the same variance) are independent zero-mean Gaussians with

$$\sigma_U^2 = \frac{\sigma_\ell^2(D)}{c^2} [1 - C_\alpha(\tau; D)] (1 + \Phi), \quad \sigma_V^2 = \frac{4\sigma_\ell^2(D)}{c^2} [1 - C_\alpha(\tau; D)] (1 - \Phi). \quad (\text{A14})$$

In the narrow-band regime $\sigma_V \ll \tau_L$, so $V_{\text{src}}(\tau - U - V/2) V_{\text{src}}(\tau - U + V/2) \approx V_{\text{src}}^2(\tau - U)$, and the V integral is the characteristic function of a Gaussian:

$$\int dV \mathcal{N}(V; 0, \sigma_V^2) e^{-i\omega_0 V} = e^{-\omega_0^2 \sigma_V^2 / 2} = \exp\left(-2\sigma_\phi^2(D) [1 - C_\alpha(\tau; D)] [1 - \Phi]\right), \quad (\text{A15})$$

with $\sigma_\phi = \omega_0 \sigma_\ell / c$ the rms accumulated carrier phase along a single ray. The result reproduces Eq. (43) of the body, with the cross-direction carrier-phase factor $\mathcal{F}_{\text{cross}}(\tau; b)$ of Eq. (44).

7. Summary

Equations (40) and (43) of the body are the key results of this pedestrian derivation. Two structural features deserve emphasis: a cross-direction carrier-phase factor $\mathcal{F}_{\text{cross}}$ controlled by the imaging-channel scale σ_ϕ appears (so the $\sigma_\phi/\sigma_\Delta/\tau_c$ decoupling of Section III F 2 is exact only for a point source); and the envelope-convolution variance is $\sigma_U^2 = (\sigma_\ell^2/c^2)[1 - C_\alpha](1 + \Phi)$ rather than the point-source $\sigma_\Delta^2 = (2\sigma_\ell^2/c^2)[1 - C_\alpha]$, the two coinciding only at $\Phi = 1$. At $\tau = 0$ both $\mathcal{F}_{\text{cross}}$ and σ_U^2 trivialize ($C_\alpha(0) = 1$), reproducing the rigorous single-time identity $g_{\text{obs}}^{(2)}(0) = 2$ for any $\Phi_{\text{eff}} \in [0, 1]$.

-
- [1] J. A. Wheeler, *Physical Review* **97**, 511 (1955).
 - [2] S. W. Hawking, *Nuclear Physics B* **144**, 349 (1978).
 - [3] S. Carlip, *Rep. Prog. Phys.* **86**, 066001 (2023), arXiv:2209.14282 [gr-qc].
 - [4] Y. J. Ng and E. S. Perlman, *Universe* **8**, 382 (2022).
 - [5] Y. J. Ng, *Physical Review Letters* **86**, 2946 (2001).
 - [6] Y. J. Ng, *Modern Physics Letters A* **18**, 1073 (2003).
 - [7] R. Lieu and L. W. Hillman, *The Astrophysical Journal* **585**, L77 (2003).
 - [8] W. A. Christiansen, Y. J. Ng, D. J. E. Floyd, and E. S. Perlman, *Physical Review D* **83**, 084003 (2011).
 - [9] E. S. Perlman, S. A. Rappaport, W. A. Christiansen, Y. J. Ng, J. DeVore, and D. Pooley, *The Astrophysical Journal* **805**, 10 (2015).
 - [10] V. S. H. Lee, K. M. Zurek, and Y. Chen, *Physical Review D* **109**, 084005 (2024), arXiv:2312.06757 [gr-qc].
 - [11] R. H. Brown and R. Q. Twiss, *Nature* **177**, 27 (1956).
 - [12] A. J. F. Siegert, *On the fluctuations in signals returned by many independently moving scatterers*, Tech. Rep. 465 (MIT Radiation Laboratory, 1943).
 - [13] W. Martienssen and E. Spiller, *American Journal of Physics* **32**, 919 (1964).
 - [14] J. W. Goodman, *Speckle Phenomena in Optics: Theory and Applications* (Roberts and Company Publishers, Englewood, Colorado, 2007).
 - [15] L. Mandel and E. Wolf, *Optical Coherence and Quantum Optics* (Cambridge University Press, Cambridge, 1995).
 - [16] A. U. Abeysekara *et al.*, *Nature Astronomy* **4**, 1164 (2020).
 - [17] A. Acharyya *et al.*, *The Astrophysical Journal* **966**, 28 (2024).
 - [18] The Maréchal formula $S = e^{-\sigma_\phi^2}$ (valid for $\sigma_\phi \lesssim 1$) and the temporal result $\mathcal{F}_{\text{foam}}(\infty) = e^{-\sigma_\phi^2}$ are linked by a spatio-temporal ergodic identification: both require $r_\perp \ll D_{\text{tel}}$ and $C_\alpha(\infty; D) \rightarrow 0$. For quasi-static foam ($C_\alpha \approx 1$) or $r_\perp \gg D_{\text{tel}}$, the aperture sits within a single coherence patch, the wavefront carries only a common phase offset, and $S \rightarrow 1$ even when σ_ϕ^2 is large. The transverse coherence function $\Phi(b/r_\perp)$ governs a physically distinct effect — cross-ray correlations at angular separations $\theta \neq \theta'$ — which enters the second-order coherence of extended sources (Section IV) but not the single-aperture Strehl ratio.
 - [19] R. Ragazzoni, M. Turatto, and W. Gaessler, *ApJ* **587**, L1 (2003), arXiv:astro-ph/0303043 [astro-ph].
 - [20] A. A. Abdo, M. Ackermann, M. Ajello, *et al.* (Fermi GBM/LAT Collaboration), *Nature* **462**, 331 (2009), arXiv:0908.1832 [astro-ph].
 - [21] F. Aharonian *et al.* (H.E.S.S. Collaboration), *Phys. Rev. Lett.* **101**, 170402 (2008), arXiv:0810.3475 [astro-ph].
 - [22] LHAASO Collaboration, *Physical Review Letters* **133**, 071501 (2024), arXiv:2402.06009.
 - [23] C. Foellmi, *Astronomy & Astrophysics* **507**, 1719 (2009).
 - [24] D. Dravins, S. LeBohec, H. Jensen, and P. D. Nuñez, *Astroparticle Physics* **43**, 331 (2013).
 - [25] W. Guérin, A. Dussaux, M. Fouché, G. Labeyrie, R. Donnerstein, D. Nakar, A. Gontier, and R. Kaiser, *Monthly Notices of the Royal Astronomical Society* **472**, 4126 (2017).
 - [26] W. Guérin, M. Hugbart, S. Tolila, N. Matthews, O. Lai, J.-P. Rivet, G. Labeyrie, and R. Kaiser, *Comptes Rendus Physique* **26**, 659 (2025).
 - [27] A. G. Kim, P. E. Nugent, X. Chen, L. Wang, J. T. O'Brien, and Tardis Collaboration, *Phys. Rev. D* **111**, 083047 (2025), arXiv:2503.07725 [astro-ph.IM].
 - [28] Planck Collaboration, N. Aghanim, Y. Akrami, M. Ashdown, J. Aumont, C. Baccigalupi, *et al.*, *Astronomy & Astrophysics* **641**, A6 (2020), arXiv:1807.06209.
 - [29] W. Kollatschny and M. Zetzl, *Astronomy & Astrophysics* **549**, A100 (2013).
 - [30] A. Rodríguez-Ardila and X. Mazzalay, *Monthly Notices of the Royal Astronomical Society* **357**, 220 (2006).
 - [31] K. M. Leighly, J. P. Halpern, E. B. Jenkins, and D. Casebeer, *Astrophysical Journal Supplement Series* **173**, 1 (2007), arXiv:0705.0940.
 - [32] D. J. Hillier, K. Davidson, K. Ishibashi, and T. Gull, *The Astrophysical Journal* **553**, 837 (2001).
 - [33] N. Smith, J. E. Andrews, A. Rest, F. B. Bianco, J. L. Prieto, T. Matheson, D. J. James, R. C. Smith, G. M. Strampelli, and A. Zenteno, *Monthly Notices of the Royal Astronomical Society* **480**, 1466 (2018).
 - [34] F. Taddia, M. D. Stritzinger, J. Sollerman, M. M. Phillips, J. P. Anderson, L. Boldt, A. Campillay, S. Castellón, C. Contreras, G. Folatelli, S. M. Haberman, M. Hamuy, E. Heinrich-Josties, E. Y. Hsiao, W. Krzeminski, N. Morrell, C. R.

- Burns, W. L. Freedman, B. F. Madore, S. E. Persson, M. Roth, N. B. Suntzeff, P. A. James, and G. Folatelli, *Astronomy & Astrophysics* **555**, A10 (2013).
- [35] C. L. Ransome and V. A. Villar, *The Astrophysical Journal* **987**, 13 (2025), arXiv:2409.10596 [astro-ph.HE].
 - [36] F. B. Davies, E. Bañados, S. E. I. Bosman, A. Ganguly, S. Belladitta, J. Power, and J. Rees, arXiv e-prints (2026), arXiv:2604.13152 [astro-ph.GA].
 - [37] J. W. Sulentic, P. Marziani, R. Zamanov, R. Bachev, M. Calvani, and D. Dultzin-Hacyan, *The Astrophysical Journal* **566**, L71 (2002).
 - [38] S. Zamfir, J. W. Sulentic, P. Marziani, and D. Dultzin, *Monthly Notices of the Royal Astronomical Society* **403**, 1759 (2010).
 - [39] R. Brouri, A. Beveratos, J.-P. Poizat, and P. Grangier, *Optics Letters* **25**, 1294 (2000).
 - [40] N. Smith, W. Li, A. V. Filippenko, and R. Chornock, *Monthly Notices of the Royal Astronomical Society* **412**, 1522 (2011), arXiv:1006.3899.
 - [41] S. M. Adams, C. S. Kochanek, J. F. Beacom, M. R. Vagins, and K. Z. Stanek, *Astrophysical Journal* **778**, 164 (2013), arXiv:1306.0559.
 - [42] E. P. Verlinde and K. M. Zurek, *Physics Letters B* **822**, 136663 (2021), arXiv:1902.08207 [gr-qc].